

Master Thesis

Fiete Scheel

Automated Translation into Simple German

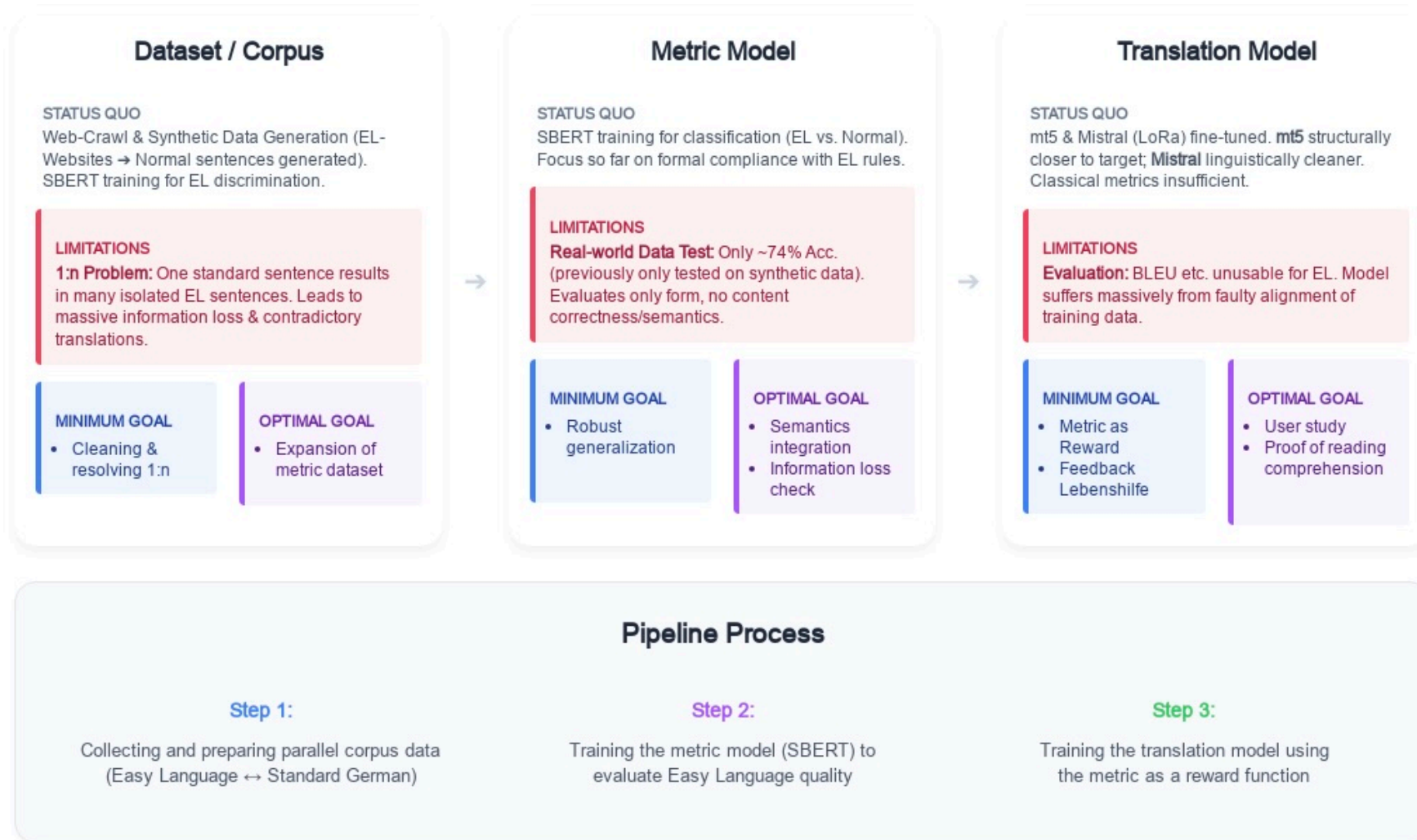
Language as a barrier: Complex texts exclude people from information and social participation.

High effort: To date, the creation of simple german texts has been almost exclusively manual and time-consuming.

Research gap:

Lack of robust, data-driven methods for distinguishing between standard and simple language in German.

Lack of automated metrics for the qualitative assessment of simplifications.



Components of this work

Week 1

Data Research

Problem:

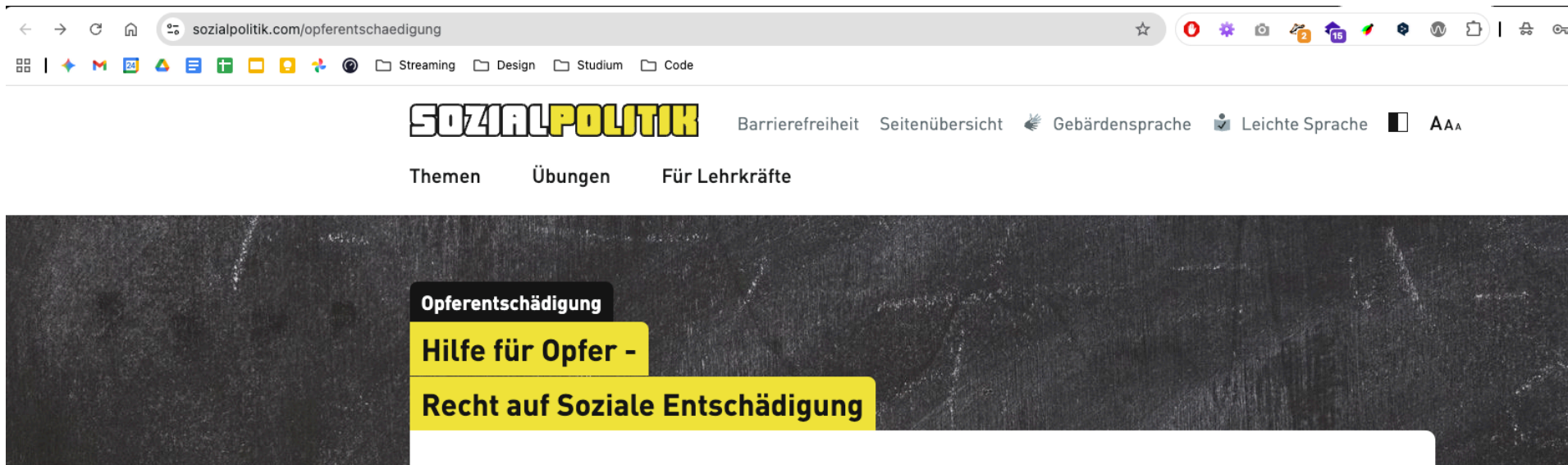
It is often very difficult to find exact parallel texts (such as news articles or blog posts) on the internet that are written in standard language (or "complex language") and simple language, and that can be systematically matched with one another.

sozialpolitik.com (Sloution with URL)

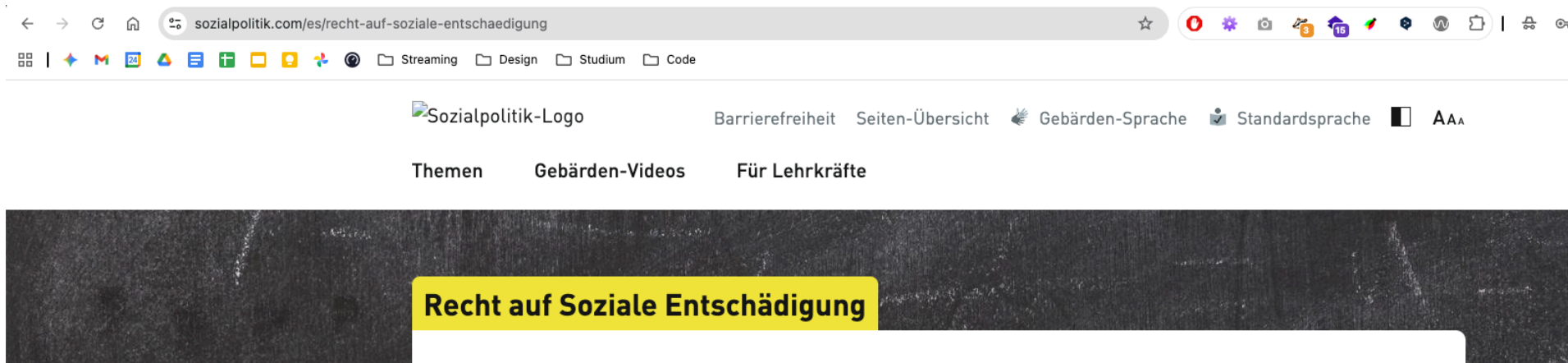
Here, the blog posts have the same title in both language versions.

The "Easy Language" version is easy to identify and access, as the URL simply has the **prefix** "ls-" added to the beginning.

While the URLs for the blog posts differ, there is a **button** on each page that allows you to **switch between** "Easy Language" and "Standard Language."



sozialpolitik.com (normal german)

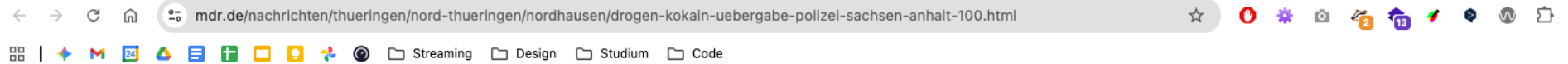


sozialpolitik.com (simple german)

At Mitteldeutscher Rundfunk, the links aren't embedded in the URL, but the user guidance within the text is excellent.

Below every article in Easy Language, there is a **direct link** to its counterpart in standard language and **vice versa**.

Not every article in standard language is translated into Easy Language, but **every article in Easy Language is translated into standard language** (as verified by random sampling).



Fast alle Verdächtigen kommen den Angaben nach aus Thüringen. Sie wurden bei dem Drogengeschäft von der Polizei überrascht und festgenommen. Laut Polizei wurden mehr als ein Kilogramm Kokain und eine Schusswaffe beschlagnahmt. Das Kokain hat einen Straßenverkaufswert von mindestens 60.000 Euro.

 ARTIKEL HÖREN

 TEILEN

Die Festnahmen sind das Ergebnis von mehreren Einsätzen der Kriminalpolizei im Großraum Sangerhausen und an mehreren Orten in Nordthüringen, hieß es. Dabei ging es der Polizei zufolge um Ermittlungen wegen des Verstoßes gegen das Betäubungsmittelgesetz.



 **BARRIEREFREIHEIT**

**Hier können Sie diese Nachricht auch in
Leichter Sprache lesen**

mdr.de (normal german)

 TEILEN

Fünf Menschen bei Kokainübergabe festgenommen - Polizeieinsatz in Thüringen und Sachsen-Anhalt

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Unfortunately, Tagesschau stands out negatively when it comes to integrating standard and simplified language, as it lacks a user-friendly and systematic structure:

- **Lack of links:** There is no button or direct link within the articles to easily switch back and forth between the different language levels.
- **Inconsistent URL structure:** While the basic structure of the URLs appears well-thought-out, in practice the articles are named inconsistently. There is no reliable pattern (such as a fixed prefix), making a systematic or even automated mapping of the texts impossible.
- **High manual effort:** To find parallel texts, the only option is a time-consuming manual search for the semantically matching counterpart.
- **Significant reduction in content:** Even if you have found the corresponding pair of articles, a direct text comparison is difficult because the version in Easy Language has been drastically shortened and greatly simplified in terms of content.

Next Steps

Researching additional sources: Searching for other news portals or official government websites (e.g., bpb.de) that systematically provide LS texts.

Quantification of sources: Determining the approximate number of available article pairs on mdr.de and sozialpolitik.com .

Scraper development: Creating a prototype for the automated extraction of articles from sozialpolitik.com (using URL logic).

MDR crawler: Investigation into whether the links between LS and standard language on mdr.de can be systematically crawled.

Matching strategy for Tagesschau: Evaluation of whether semantic similarity analyses (e.g., via embeddings) can be used to automatically establish the missing links in Tagesschau.

Week 2-3

Evaluation of institutional and governmental websites.

Observations: Significant availability of "Easy Language" (Leichte Sprache) sections, but alignment Issue.

Alignment Issue: Most institutional pages provide summaries or simplified overviews rather than 1:1 translations of standard articles.

Terminological Inconsistency: Frequent overlap and confusion between "Plain Language" (Einfache Sprache) and "Easy Language."

Saarländischer Rundfunk (SR)

Problem: Standard language content is often video/audio (broadcast format), while the Easy Language version is a brief text summary.

Impact: Automated 1:1 alignment is difficult due to the format mismatch.

Technical Workaround: Using metadata (Open Graph/OG-Images) to verify if two different formats refer to the same news event + Date of Publishment

Media Portals with High Alignment Potential

MDR (Mitteldeutscher Rundfunk): High suitability for scraping due to clear XML sitemaps and explicit "linking mechanisms" in the HTML structure.

taz: Alignment is present but less systematic; links are often embedded manually in italicized paragraphs at the end of articles with word "hier" .

Conclusion: Media portals show significantly higher potential for 1:1 alignment than administrative/government portals.

Terminological Anchors: Dictionaries



Objective: Establishing a "Ground Truth" for specific terminology.

Resources Analyzed:

- **Hurraki:** A specialized Wiki for Easy Language.
- **Nachrichtenleicht:** A dictionary section providing definitions for complex terms.

Benefit: These provide direct 1:1 word/phrase mappings, essential for terminological consistency.

Literature Review: Existing Corpora (Status 2023)

Analysis of Klaper et al. (2013)

Pros: High-quality 1:1 alignment; approx. 70,000 tokens.

Cons: "Source decay"---many URLs from 2013 are no longer active.

Next Step: Evaluating the reconstruction of this corpus using the Wayback Machine (Internet Archive).

Literature Review: Existing Corpora (Status 2023)

A New Aligned Simple German Corpus (ACL 2023)

Analysis: Investigating provided crawler scripts and alignment logic.

Key Insight: The project moves away from generic scraping toward source-specific extraction heuristics to ensure high-quality sentence alignment.

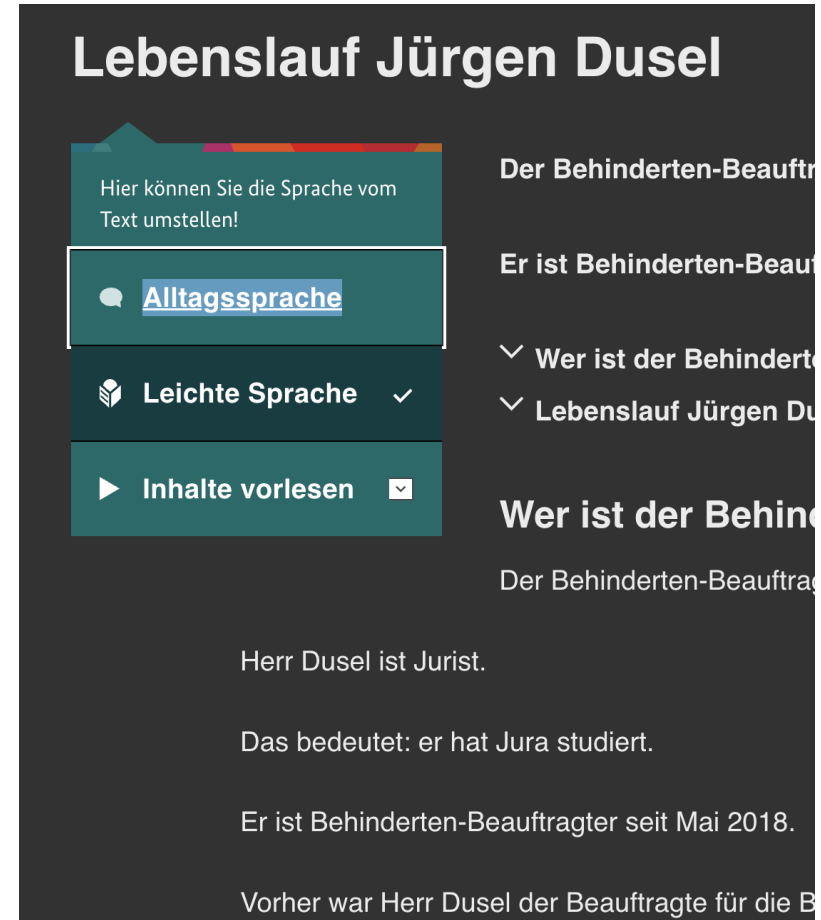
Extraction Strategies

Apotheken Umschau: Targeting specific link titles
`title="hier"` for back-references.

Behindertenbeauftragter: Identifying language
switches via CSS class patterns `.c-language-`
`switch`.

Sozialpolitik.com: Filtering for links with the class
`underline easy` and specific `hreflang` attributes.

```
<a href="/es/die-arbeits-welt" hreflang="de-DE" class="underline easy">
  Leichte Sprache
</a>
```



The screenshot shows a dark-themed website interface. At the top, the title 'Lebenslauf Jürgen Dusel' is displayed. Below the title, there is a language selection menu with three options: 'Alltagssprache' (highlighted with a blue box), 'Leichte Sprache' (with a checkmark), and 'Inhalte vorlesen' (with a play button icon). To the right of the menu, there is a sidebar with navigation links: 'Der Behinderten-Beauftragter', 'Er ist Behinderten-Beauftragter', 'Wer ist der Behinderten-Beauftragter', and 'Lebenslauf Jürgen Dusel'. Below the sidebar, there is a section titled 'Wer ist der Behinderten-Beauftragter' with a sub-header 'Der Behinderten-Beauftragter'. The main content area contains text about Jürgen Dusel, including 'Herr Dusel ist Jurist.', 'Das bedeutet: er hat Jura studiert.', 'Er ist Behinderten-Beauftragter seit Mai 2018.', and 'Vorher war Herr Dusel der Beauftragte für die B...'.

Extraction Strategies

Brand Eins: Unique case where both languages exist on the same URL; differentiation achieved through CSS color coding (Red text = Easy Language).

Stadt Köln: Using exact string matching for navigational links ("Diese Seite in Alltags-Sprache lesen").

Summary: Technical research confirms a "one-size-fits-all" scraper is ineffective; custom logic for each domain is required.

Conclusion & Next Steps

Current Activity: Development of a quantification script to scan identified sources and calculate current token counts.

Pipeline Integration: These counting scripts serve as "pre-scrapers" to validate data density before full extraction.

Literature & Expansion: Systematic review of remaining papers from the "Existing Corpora (Status 2023)" list. Identification of additional sources and site-specific scraping mechanisms.

Goal: Adaptation of the Toborek et al. crawler logic to the newly identified structures (MDR, taz, etc.).

Week 4

Initial Scraping for Token Count

Goal: Create a diverse, multi-domain corpus exceeding current benchmarks in quality and thematic breadth.

The Pipeline:

Discovery: Crawling target websites for "Simple Language" (LS) sections.

Alignment: Matching LS articles to their "Standard Language" (AS) originals.

Extraction: Pulling raw content for initial analysis.

Target Scope: News, Health, Administration, Social Policy, and Economy.

Results: Media & News

MDR (Mitteldeutscher Rundfunk):

298 LS articles scanned → 242 successfully aligned.

Largest news source in the corpus (~300k tokens total).

taz (die tageszeitung):

Small but high-quality subset (7 pairs aligned).

Challenge: One LS text often summarizes multiple AS articles.

Brand Eins:

90 articles scanned → 34 pairs aligned.

Focus on economic and social narratives.

Results: Public Sector & Administration

Hamburg.de:

156 articles scanned → 155 initial pairs.

Broad coverage of city services and regulations.

Stadt Köln:

95 articles scanned → 57 pairs aligned.

Historical data recovered from archives.

Behindertenbeauftragter:

95 articles scanned → 73 pairs aligned.

Policy-focused content with technical terminology.

Results: Specialized Topics

Apotheken Umschau (Health/Medical):

484 articles scanned → 161 pairs aligned.

Critical source for medical term simplification.

Sozialpolitik.com:

22 articles → 22 pairs (100% success rate).

Educational content on the German social system.

Lebenshilfe Main-Taunus:

59 articles scanned → 46 pairs aligned.

Community-level communication and event reporting.

Global Scraping Statistics

Metric	Result
Total Scanned LS Articles	1,347
Successfully Aligned Pairs	797
Total Tokens (Simple)	411,540
Total Tokens (Standard)	676,903
Average Success Rate	~59%
Length Ratio (AS:LS)	1.65 : 1

Tokens: words and punctuation

Findings & Lessons Learned

Content Divergence: Standard articles are significantly longer, suggesting that simplification involves heavy summarization, not just sentence splitting.

Alignment Gaps: 41% of simple language articles lack a direct 1:1 standard counterpart on the same site.

Scale: With 797 pairs, the new corpus is already larger than most existing document-aligned German datasets.

Week 5

Focus: From Quantity to Quality

The Noise Problem: Raw scraping produced significant "boilerplate" (menus, ads, legal footers).

Implement source-specific minimal scripts to ensure that the tokens in the dataset are strictly editorial content (not menus, ads etc.).

Case Study: Apotheken Umschau

Issue: Figcaptions ("The image shows..."), internal table of contents (TOC), and "Sign up now" banners were polluting the text.

Solution:

Decomposition: Stripped <figcaption>, <figure>, and .copyright elements.

TOC Filter: Identified and removed blocks containing only internal anchor links.

Blacklist: Regex filters for "Das Bild zeigt" and registration prompts. (extend in future)

Case Study: Brand Eins

Issue: AS and LS content lived in the same HTML blocks; simple paragraph splitting failed due to irregular formatting.

Solution:

Deep-Color Inspection: Scraper now analyzes inline CSS for the red color codes (#ff0000) used by Brand Eins to highlight simple language.

Tag-Based Extraction: Using and span-styles as semantic markers for language levels rather than just structure.

Case Study: Hamburg.de

Issue: Many articles were machine-translated (MT), marked by a "Computer has translated this" disclaimer. These are unsuitable for a gold-standard corpus.

Solution:

MT-Detection: Automated scanning for MT-disclaimers; immediate exclusion of affected pairs.

Precision Alignment: Restricted search for AS-links to the official .km1-language-bar, reducing "Ghost Alignments" from 155 down to 57 high-quality pairs.

Case Study: MDR

Issue: "The Echo Effect" - Nested `<div\>` and `<p\>` tags caused the same text to be extracted twice.

Solution:

Parent-Check Algorithm: Elements are only extracted if none of their ancestors are already in the extraction list.

Idea: Length-Ratio Filter: Pairs with a ratio > 5.0 (e.g., a Live-Ticker vs. a short summary) are automatically discarded as "Unbalanced."

Case Study: Archival Sources

Cities of Cologne & Main-Taunus:

Encoding Issues: Resolved "Umlaut" errors (e.g., kÃ¶nnen) by forcing response.apparent_encoding.

Redundancy: Content hashing introduced to prevent duplicate articles caused by multiple URL parameters (m-20, m-79) pointing to the same archive snapshot.

Truncation: Automatically cutting off repetitive contact blocks ("Ansprechpartner:", "Telefon:").

Automated Data Cleaning Pipeline

HTML De-cluttering: Remove non-article tags (nav, aside, footer).

Structural Extraction: Map headers, lists, and paragraphs with proper spacing (\n).

Regex Filtering: Strip author credits, donation calls (taz), and radio-station promos (MDR).

Identity Filter: Compare LS and AS strings; discard pairs where the content is identical (common in news sites).

Deduplication: Hash-based check for uniqueness across the entire corpus.

Resulting Quality Improvement

Cleaner Data: Average noise reduction of 15-20% per article.

Better Alignment: Improved thematic coherence by filtering out "Hub-pages" and "Tag-overviews."

Final Count: A refined, manually verified collection of ~600-700 high-quality pairs (post-filtering).

Zwischenergebnis Corpus

Source	LS Tokens	AS Tokens	Total
apotheken	147,539	249,790	397,329
behindertenbeauftragter	27,822	34,113	61,935
brandeins	7,275	7,423	14,698
hamburg	44,358	40,063	84,421
koeln	40,868	24,182	65,050
main_tanus	7,289	6,715	14,004
mdr	70,761	100,608	171,369
sozialpolitik	6,654	14,651	21,305
taz	4,589	8,027	12,616

Summary of tokens (words and punctuation)

Next Steps

Apotheken Umschau: Investigation into why so few URLs from this source could be successfully aligned (root cause analysis for low alignment rate).

Length Ratios: Analysis of discrepancies between Plain Language (PL) and Everyday Language (EL) across various sources. Why do some sources systematically have more PL tokens than EL tokens, and others the reverse?

Historical data acquisition: Evaluation of the Wayback Machine to achieve broader coverage of articles from different time periods (past and present) and thus expand the corpus.

Template Script: Because of redundant code, build a template script with core functionalities

Week 6

Weekly Focus: City and State Portals

- **Strategy:** Systematic review of states and state capital portals (Stuttgart, Wiesbaden, Hannover).
- **Findings:**
 - High variance in the implementation of legal accessibility requirements.
 - Hannover identified as the absolute "top performer".
 - All sites have the same entrypoint page for "Leichte Sprache"

Legal Obligations

Why do many government sites offer so little parallel data?

1. **BITV 2.0 Minimum Solution:** The law only requires LS for:

- Essential tasks (homepage)
- Navigation & Accessibility statements

2. **No Full-Text Obligation:** News or technical articles do not legally have to be translated.

Case Study: Hannover.de (The Gold Standard)

- **Scale:** Over 800 article pairs – the largest single source in the corpus.
- **Technical Features:**
 - **Alignment:** URL logic - the AS version is primarily derived by removing the query parameter `sp:out=easy` (or `?sp%3Aout=easy`) from the LS URL.
 - **Quality:** High coherence between Easy-to-Read (LS) and Standard Language (AS).

Technical Challenges: Surgical Cleaning

Problem: Standard extraction yields 10-20% boilerplate noise.

- *UI Fragments:* "Print page", "In sign language", "Send email".
- *Navigation:* "Back to overview", "Related topics".

Solution:

- Implementation of CSS blacklists (decomposition of containers like `.SP-Tools`, `.header-1s`).
- Regex-based filtering of repetitive standard sentences.
- **Result:** Significant increase in token precision and cleaner vocabulary statistics.

Problem: Wiesbaden Alignment

Insight: Technical alignment does not equal semantic alignment.

- **Symptom:** URL parameters suggest a translation, but content varies significantly.
- **Examples:**
 - AS: History of the city forest vs. LS: General forest rules.
 - AS: Press release on a poster campaign vs. LS: Explanation of "Smart City".
- **Action:** Introduction of a manual audit step before the model training phase. Wiesbaden data may be excluded.

Current Status (Metrics)

Source	Pairs	Words (LS)	Words (AS)	Tokens (LS)	Tokens (AS)
(...)	189	72,816	79,585	135,484	168,489
MDR (prev.)	235	53,869	83,487	98,780	173,765
Apotheken (prev.)	161	123,505	205,063	241,325	451,812
Hamburg (prev.)	57	34,124	33,688	61,455	73,137
Week 06 Additions					
Hannover	808	458,621	405,321	872,291	871,830
Stuttgart	42	23,653	46,060	45,202	106,629
Wiesbaden	41	7,138	10,127	13,808	23,332
Total	1,533	773,726	863,331	1,468,345	1,868,994

Tokens counted using the `tiktoken` library with the `cl100k_base` encoding.

Next Steps

1. **Finalize Extraction:** Complete final sources.
2. **(Ongoing):** Find more sources systematically.
3. **Train first model:** Train first model to see some first results.

Week 7

Weekly Focus: Measuring Information Loss

- **Hypothesis:** Texts in Easy Language (LS) don't just simplify syntax; they lose significant, concrete information compared to Standard Language (AS), even when they are longer due to explanations.
- **Goal:** Move beyond token counts and develop a multi-dimensional NLP methodology to quantify this information loss.
- **Corpus:** Analysis executed across the entire aligned dataset (1,501 article pairs).

Methodology: Similarity

1. Semantic Similarity (Embeddings)

- **Tool:** Sentence-BERT (`paraphrase-multilingual-MiniLM-L12-v2`).
- **How:** Calculate Cosine Similarity from AS and LS texts.
- **Why:** Captures whether the "core message" remains, even if heavily paraphrased.

2. Fact Retention via NER Recall

- **Tool:** spaCy (de_core_news_lg).
- **How:** Extract entities (Persons, Locations, Organizations) from AS and check if they survive in LS.
- **Why:** Measures hard information loss. If names and places vanish, factual detail is lost or is being described differently.

3. Linguistic & Syntactic Metrics

- **Tool:** spaCy POS-Tagging.
- **How:** Analyze Lexical Density, Average Sentence Length, and Part-of-Speech (POS) shifts (e.g., ratio of Adjectives, Conjunctions).
- **Why:** Reveals *how* the simplification is achieved structurally (e.g., dropping conjunctions implies loss of subordinate clauses / hypotaxis).

Key Finding 1: Massive Loss of Factual Entities

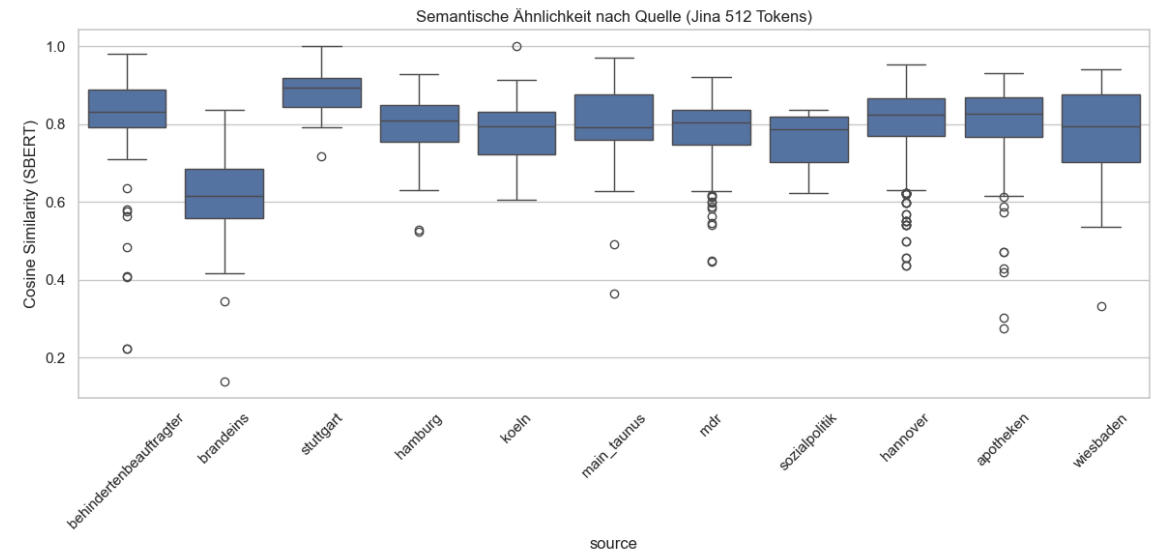
Observation: Across almost all sources, the **NER Recall is extremely low (Ø ~15-25%)**.

- This means roughly 80% of specific entities (names, locations, dates) mentioned in the Standard Language text do *not* appear in the Easy Language version.
- **Conclusion:** LS achieves simplification primarily by omitting specific facts and generalizing content, confirming a massive factual information loss.

Key Finding 2: Token Expansion vs. Semantic Shift

Observation: In some sources like **Hannover** (Ratio: 1.89) and **Köln** (Ratio: 1.83), the LS texts are nearly twice as long as the AS texts.

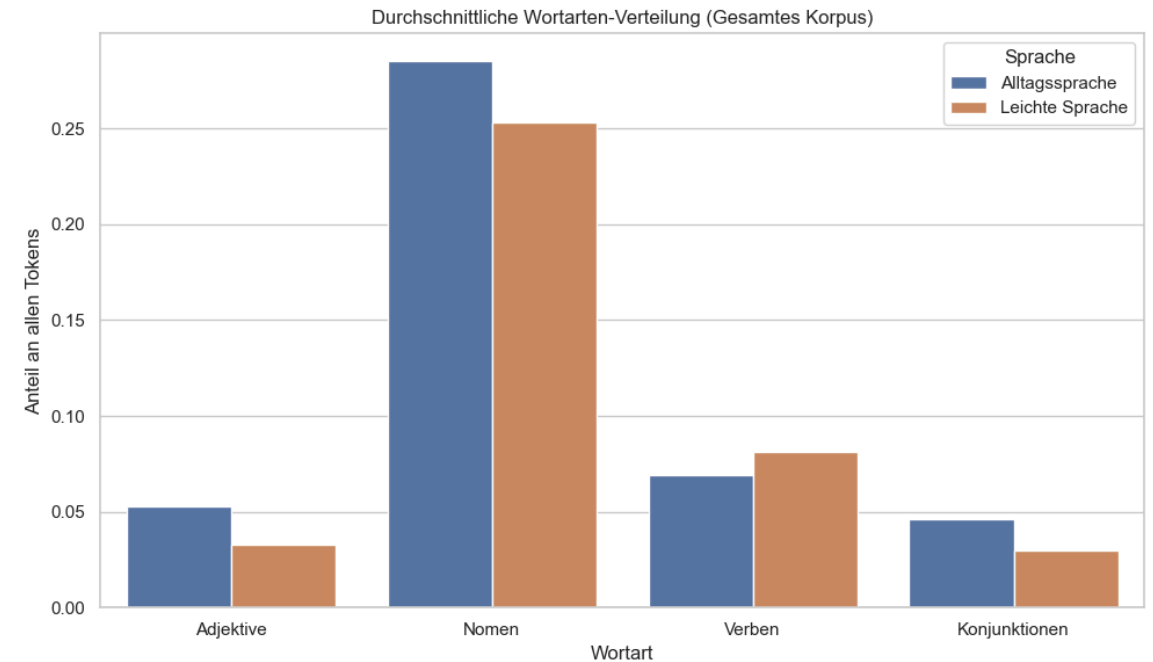
- *Why?* Complex terms are replaced with lengthy, multi-sentence explanations.
- *Result:* Despite being longer, the semantic similarity drops to ~0.70. The text explains more, but delivers less original detail.



Key Finding 3: Structural Simplification (POS)

Observation: Clear shifts in word classes between complex and simple language.

- **Conjunctions & Adjectives:** Significant decrease in **simple** language.
- **Verbs vs. Nouns:** **Simple** language uses more Verbs and fewer Nouns.
- **Stylistic Shift:** This confirms a transition from **complex** to **simple** style, which is easier to process.



Summary of Analysis Results

Source	Token Ratio (LS/AS)	NER Recall	Sem. Similarity (SBERT)
apotheken	0.98	0.09	0.64
behindertenbeauftragter	0.91	0.30	0.75
hannover	1.89	0.25	0.71
koeln	1.83	0.20	0.68
mdr	0.85	0.22	0.73
stuttgart	0.97	0.24	0.90
sozialpolitik	0.46	0.10	0.71

Extract of the full analysis results across 1,501 pairs.

Mögliche Titelvarianten & Forschungsfokus

1. Allgemein & Flexibel:

- *Automatisierte Textvereinfachung für Leichte Sprache: Ein Framework zur datengestützten Modellierung und Evaluation*

2. Ausgewogen & Methodisch:

- *Neuronale Textvereinfachung in Leichte Sprache: Entwicklung domänenspezifischer Datensätze und automatisierter Bewertungsmetriken*

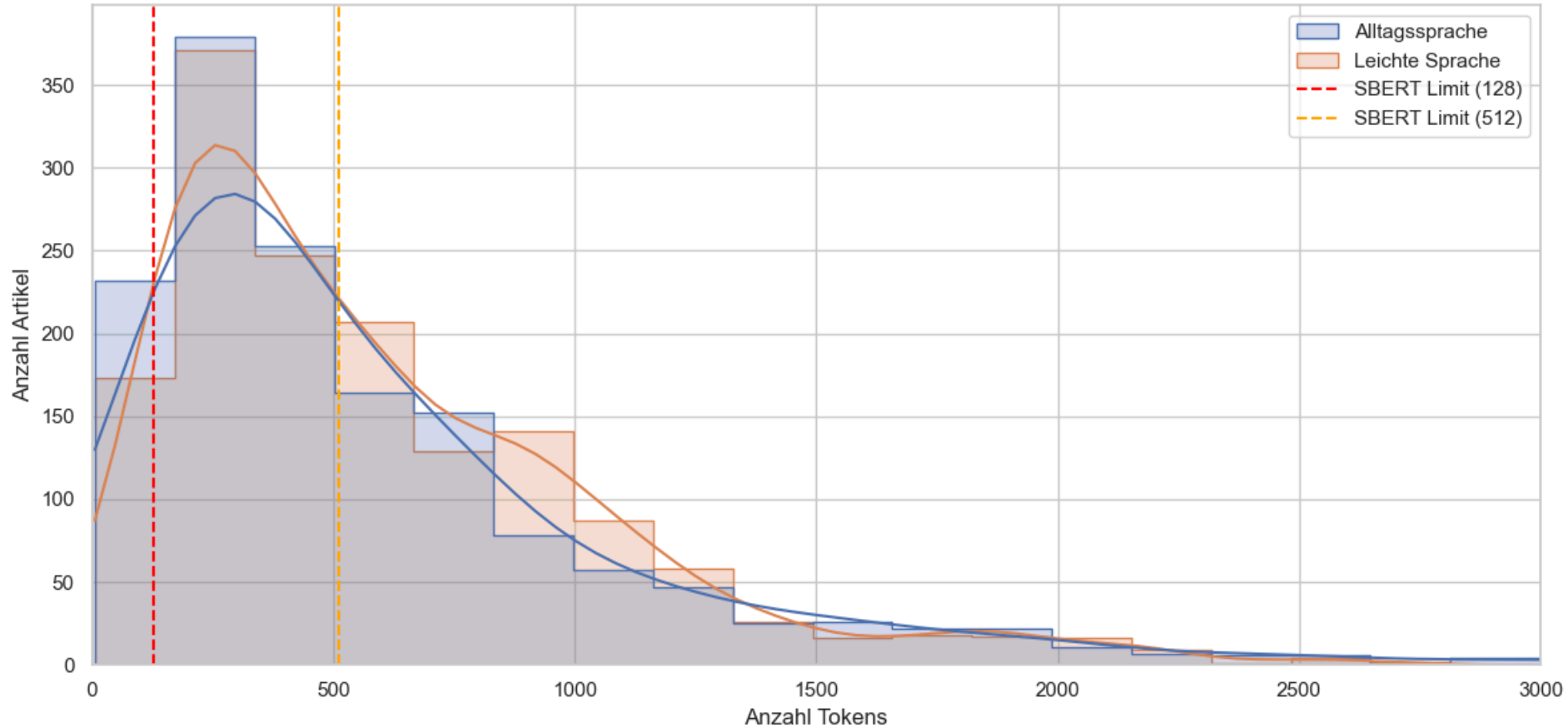
3. Spezifisch & Technisch:

- *Optimierung der maschinellen Übersetzung in Leichte Sprache: Aufbau eines Gold-Standard-Korpus und Training von Reward-Modellen zur qualitativen Modellsteuerung*

Week 8

- **Problem:** Initial SBERT analysis (Week 7) was limited to **128 tokens**, resulting in >90% of articles being truncated.
- **Goal:** Increase model coverage and validate if semantic similarity holds across full-length articles.
- **Methodology Update:**
 - Scale up to **512 tokens** (MiniLM limit).
 - Introduction of **Jina Embeddings v2** (8192 tokens) for 100% coverage.
 - **Bidirectional NER:** Analysis of "Faktentreue" (LS -> AS).

Verteilung der Artikellängen (Token-Anzahl)



Article Length Distribution

SBERT Limitation & Coverage Analysis

A statistical review of token counts revealed that the 128-token limit was capturing primarily "teasers" or "introductions," which often contain boilerplate or navigation instructions in LS.

Coverage	AS (Limit: 128)	LS (Limit: 128)	AS (Limit: 512)	LS (Limit: 512)
Fully Captured	9.4 %	5.1 %	56.9 %	52.9 %
Truncated	90.6 %	94.9 %	43.1 %	47.1 %

Lesson Learned: Even at 512 tokens, nearly half of the corpus is cut off. This necessitated the switch to a long-context model (Jina) to ensure total information retention during analysis.

Optimized Analysis: 128 vs. 512 Tokens

Increasing the context window significantly changed the measured similarity for most sources.

Source	Sim (128 Tokens)	Sim (512 Tokens)	Difference
apotheken	0.636	0.894	+0.258
hamburg	0.665	0.804	+0.139
koeln	0.684	0.833	+0.149
sozialpolitik	0.706	0.850	+0.144
wiesbaden	0.750	0.642	-0.108

Interpretation: For long texts (Apotheken, Köln), similarity rises with context. For others (Wiesbaden), it drops, indicating that while introductions are aligned, the core content diverges significantly.

Model Comparison: MiniLM vs. Jina (128 & 512 Tokens)

To ensure the model choice introduces no bias, we compared the original model (`MiniLM`) with the new one (`jina-embeddings-v2-base-de`) at identical limits.

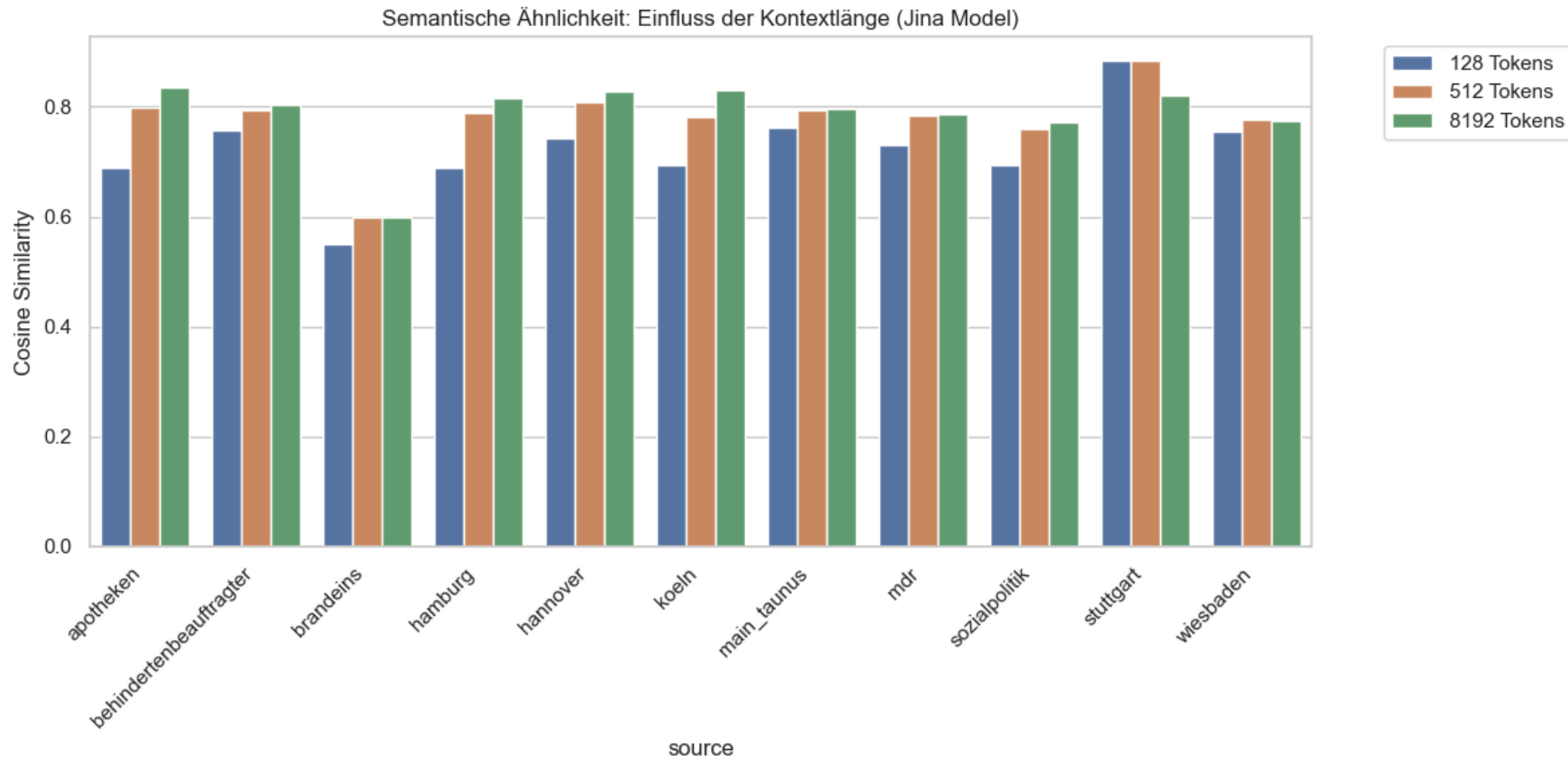
Conclusion:

- **Same Trends:** Higher context equals higher similarity in both.
- **Stability:** Jina is more stable with longer texts (e.g., Wiesbaden doesn't drop abruptly). It validates that LS stays semantically close to AS.

Source	MiniLM (128)	Jina (128)	MiniLM (512)	Jina (512)
apotheken	0.636	0.688	0.894	0.800
behindertenbeauftragter	0.746	0.756	0.761	0.793
brandeins	0.637	0.549	0.698	0.599
hamburg	0.665	0.690	0.804	0.790
hannover	0.706	0.742	0.776	0.807
koeln	0.684	0.693	0.833	0.782
main_taurus	0.762	0.763	0.727	0.794
mdr	0.733	0.731	0.766	0.784
sozialpolitik	0.706	0.694	0.850	0.760
stadtverwaltung	0.888	0.884	0.850	0.884

Solving Truncation: The Jina Model (8192 Tokens)

Source	Jina (128 Tokens)	Jina (512 Tokens)	Jina (Full / 8192)
apotheken	0.688	0.800	0.836
behindertenbeauftragter	0.756	0.793	0.804
hannover	0.742	0.807	0.828
koeln	0.693	0.782	0.829
stuttgart	0.884	0.884	0.821

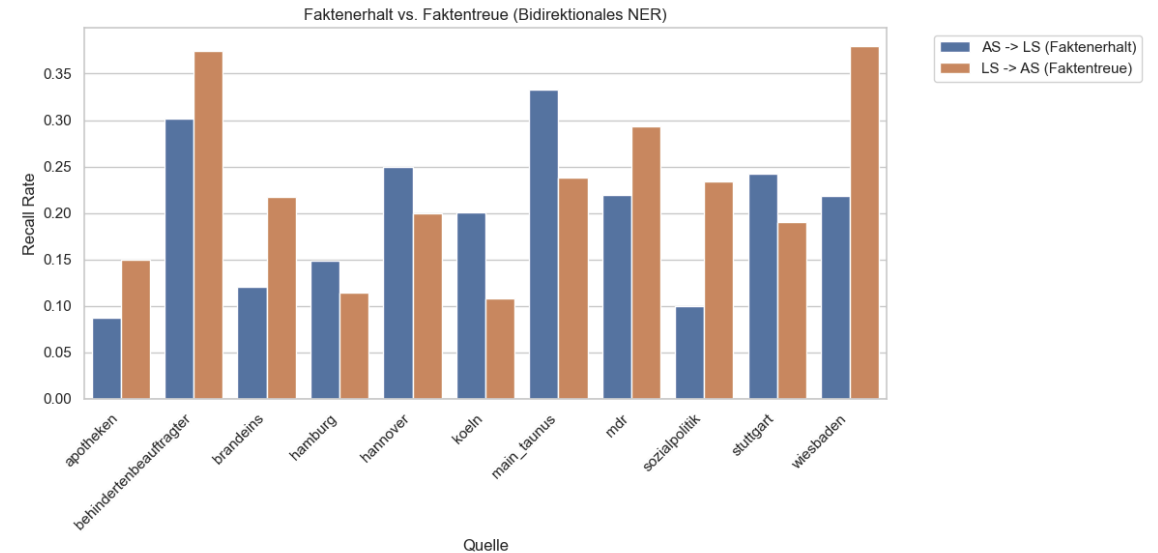


Solving Truncation: The Jina Model (8192 Tokens)

Bidirectional NER: Does LS "Invent" Facts?

We measured not just what survives (AS $\rightarrow$ LS), but also if LS introduces new entities (LS $\rightarrow$ AS).

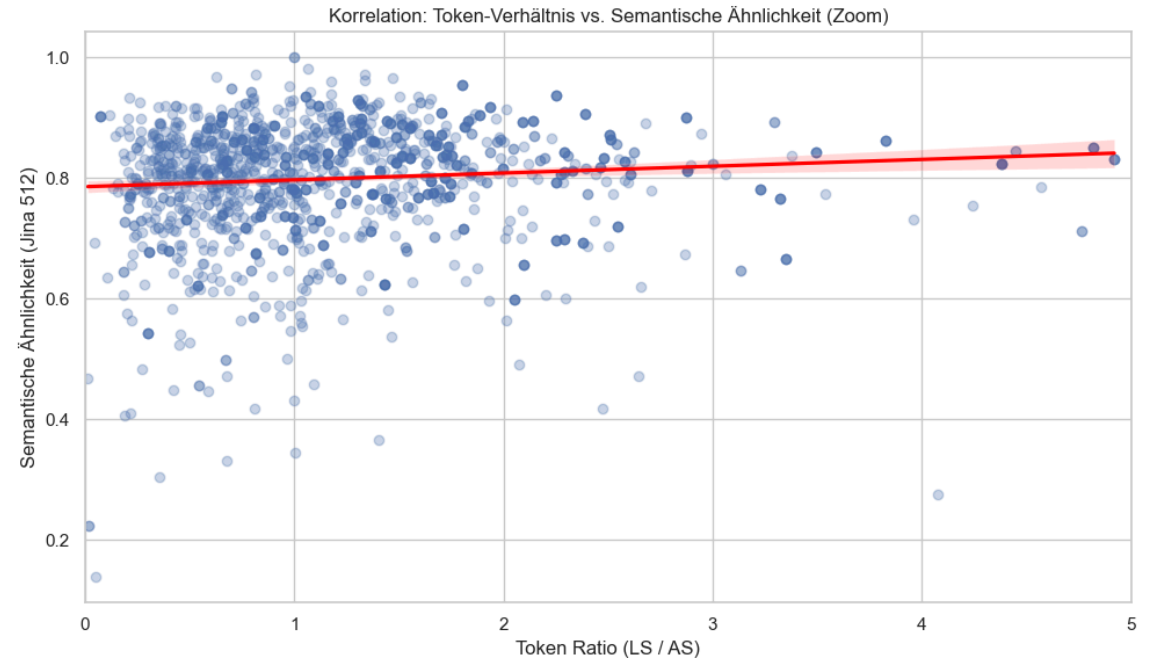
Finding: Both metrics are low. LS doesn't necessarily invent facts, but it rephrases entities into common nouns (e.g., "Arbeitsagentur" $\rightarrow$ "Amt"), which current NER models fail to align.

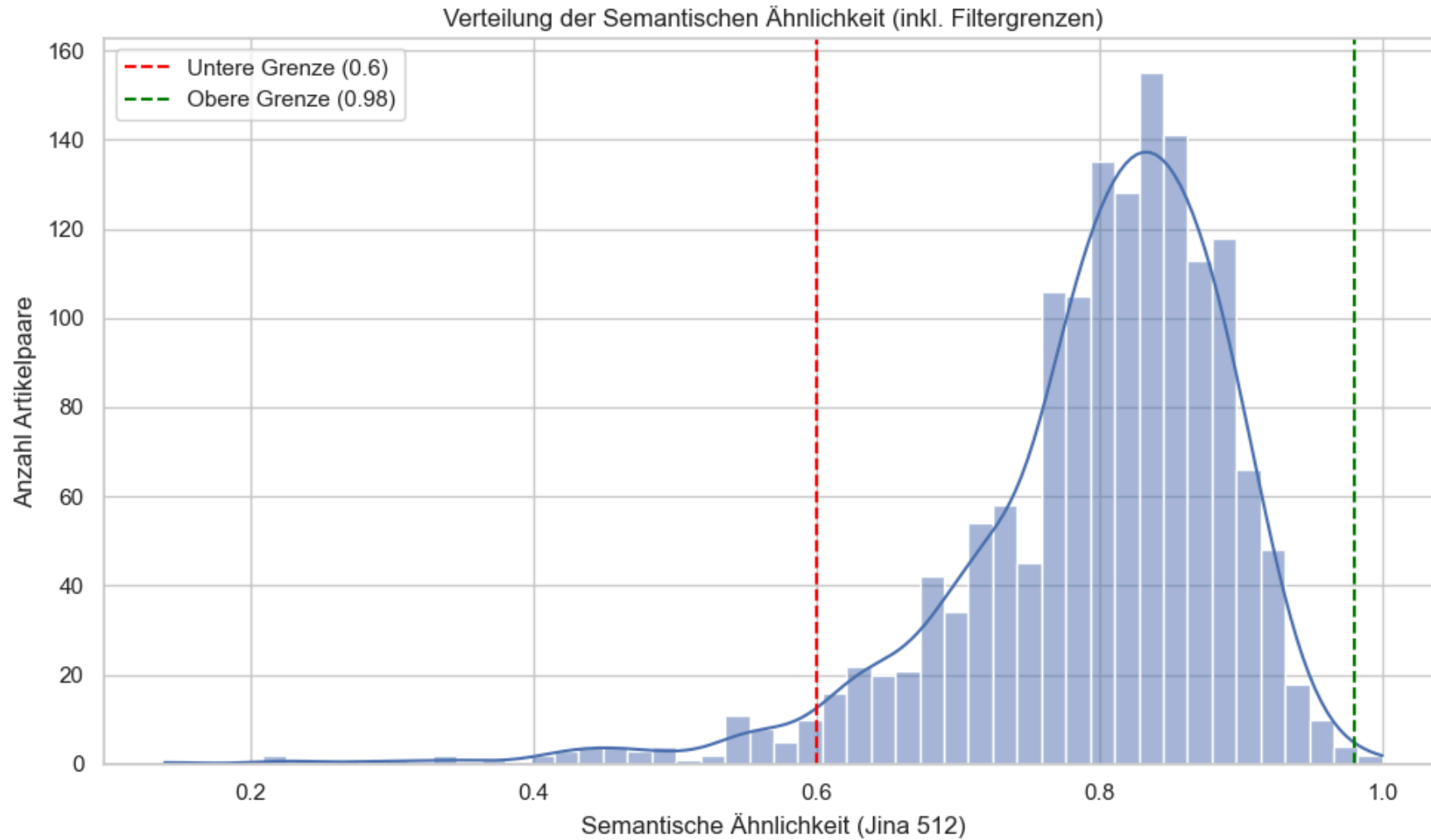


Correlation: Token-Ratio vs. Similarity

Finding: No significant linear correlation between text length expansion and semantic similarity.

Longer explanations in LS don't automatically guarantee higher semantic proximity to the original.



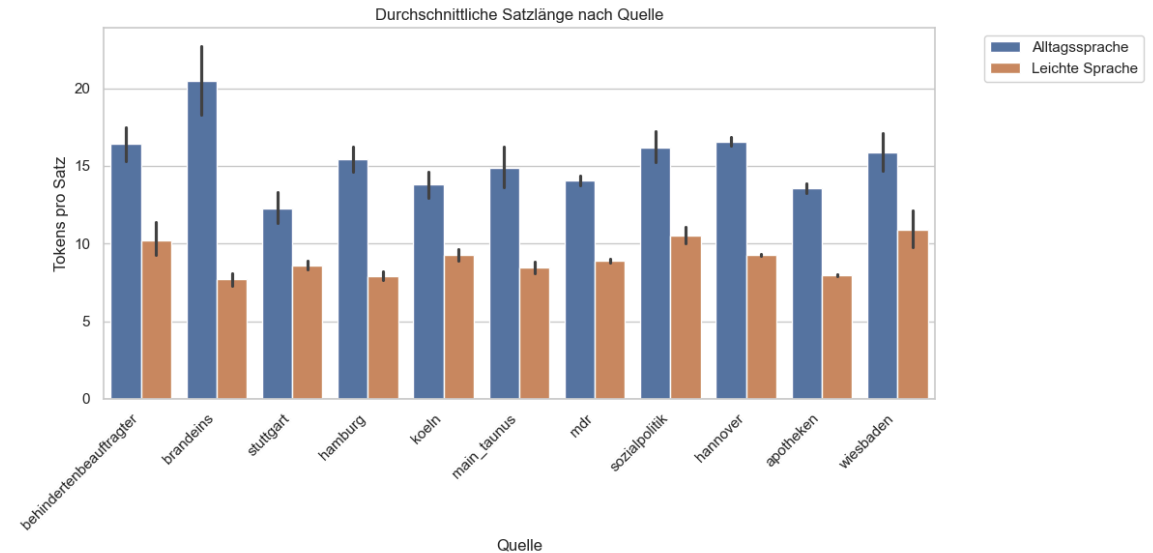


Manual Audit & Corpus Cleaning

Linguistic Shifts: Sentence Length

Analysis of 1,526 pairs confirms structural simplification.

- **Sentence Length:** Dropped from **15.6 tokens** (AS) to **9.1 tokens** (LS) – a 42% reduction.
- **Impact:** Drastic reduction in cognitive load per processing unit.



Next Steps

1. **Dataset Cleaning:** Apply the 0.6 - 0.98 similarity filter to create the "Gold Standard" training corpus. Also look for all other extremes like ratio too high etc.

Thesis-Titel

Entwicklung domänenspezifischer Datensätze und automatisierter Evaluation für ein Framework zur neuronalen Textvereinfachung in Leichte Sprache

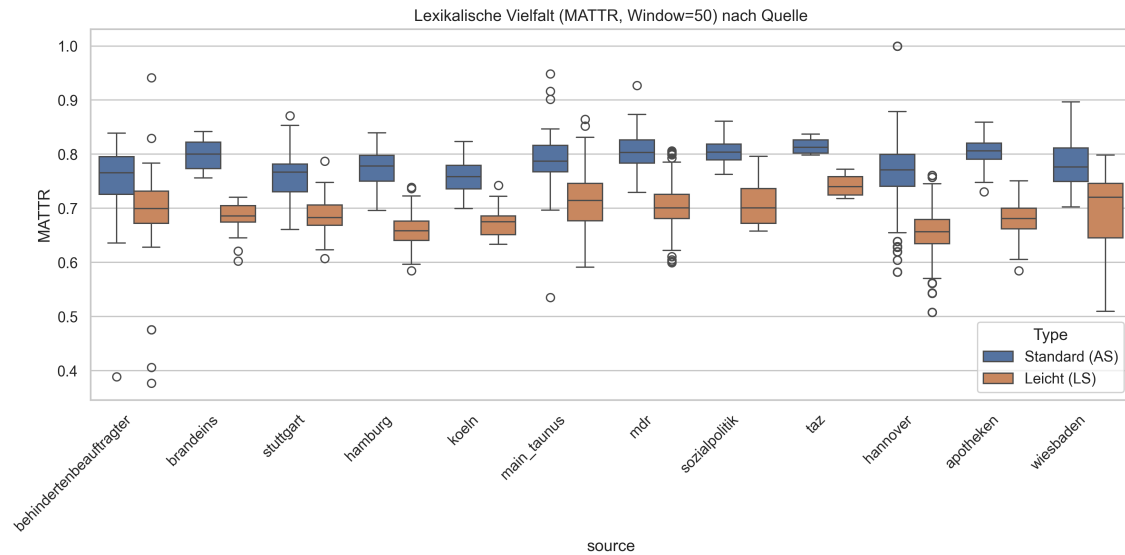
Week 11

Lexical Diversity (Type-Token-Ratio)

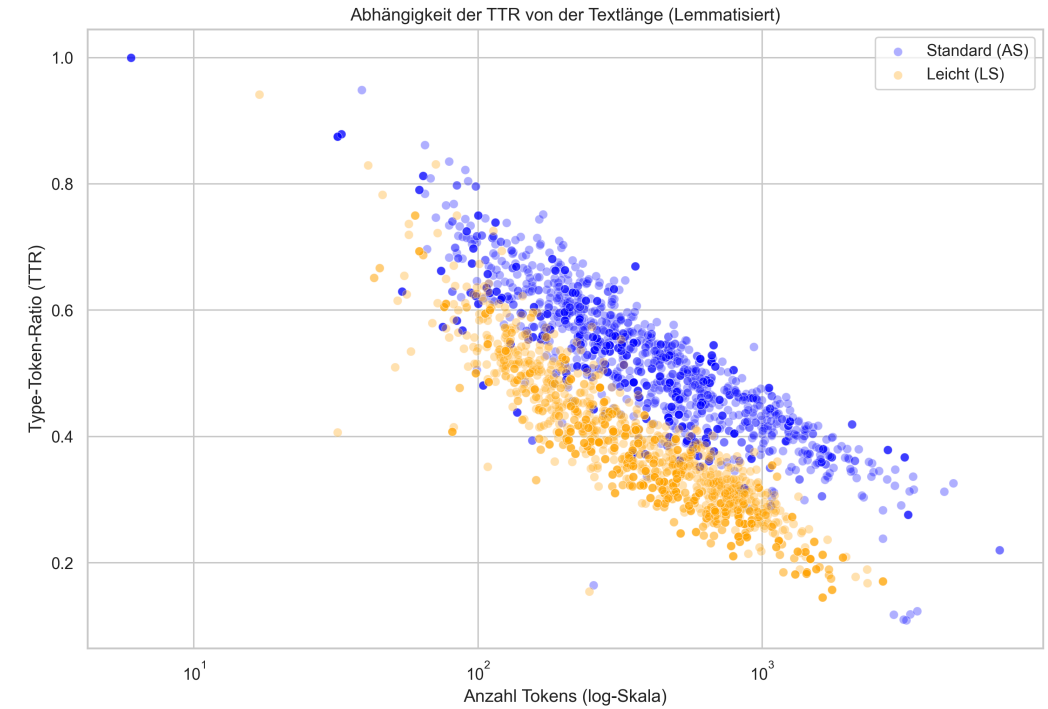
Average Reduction: The lexical variety in LS is reduced by an average of **13.6 %** compared to AS.

- **Averages:** AS (0.778) vs. LS (0.672).
- **Top Simplifiers:** *Hannover* (0.656) and *Hamburg* (0.658) show the most rigorous vocabulary reduction.
- **Journalistic LS:** The *taz* (0.742) maintains the highest lexical diversity, indicating a "higher-level" Easy Language.

Visualization of Lexical Diversity



Comparison of lexical diversity (MATTR) by source.



TTR relative to text length (log-scale) with regression lines.

Week 12

Model Training Strategy

Goal: Establish a baseline for binary classification (Normal vs. Easy German) and identify optimal data filters.

- **First Step: Baseline:** Bi-directional LSTM (BiLSTM) for lightweight comparison.

Sentence-Level Classification

Similarity Range	Balanced Accuracy
0.60 - 0.98	92.48 %
0.70 - 0.98	92.43 %
0.80 - 0.98	92.99 %
0.90 - 0.98	90.55 %

The range 0.80 - 0.98 provides the best balance between data volume and semantic alignment.

Article-Level Classification

Does full context (up to 512 tokens) improve the distinction?

Similarity Range	Balanced Accuracy
0.60 - 0.98	95.93 %
0.70 - 0.98	97.30 %
0.80 - 0.98	99.03 %
0.90 - 0.98	98.44 %

Finding: The jump from **93 % (sentence)** to **99 % (article)** confirms that "Easy Language" is a holistic stylistic phenomenon that becomes nearly perfectly distinguishable at the document level.

Training Configuration (BiLSTM)

The following hyperparameters were used to achieve the baseline performance (99% BAcc):

Parameter	Value
Optimizer	AdamW
Learning Rate	10^{-3}
Weight Decay	0.01
Batch Size	32
Max Epochs	30
Early Stopping Patience	7
Dropout	0,4
Max Seq Len	512 (Articles)

Next Steps & Research Questions

1. **Out-of-Domain Validation:** Testing the models on external data (e.g., Lebenshilfe Kiel) to check for overfitting on governmental styles.
2. **Hyperparameter Testing** Change hyperparameters systematically to find the best setup
3. **Long-Context Transformers:** Exploring models that can process up to 8,192 tokens to capture very long administrative articles without truncation.
4. **Vocabulary Pruning:** Investigating if removing rare words (< 3 occurrences) improves the classifier by forcing it to focus on common simplification patterns.

Week 13

Some Thoughts...

- **Paragraph Control Experiment:**

- filtered a paragraph-free version of the dataset (`lebenshilfe_dataset_no_paragraphs.json`).
- with `spaCy` we can filter out all whitespace tokens.

- **Vulnerability to Data Leakage Avoided:**

- Training data (governmental and news portals) and test data (internal documents, house rules, statutes from *Lebenshilfe*) are completely disjoint.
- An out-of-domain accuracy of >90% proves the BiLSTM captures genuine linguistic patterns of Easy Language rather than source-specific formatting.

- **Vocabulary Robustness:**

- Many specialized legal or administrative terms were absent from the training corpus and thus masked as `<unk>` (unknown tokens - freq ≤ 3).
- The model successfully classifies despite this, demonstrating reliance on structural features (sentence length, conjunctions, syntactic simplicity) rather than rote-memorizing vocabulary.

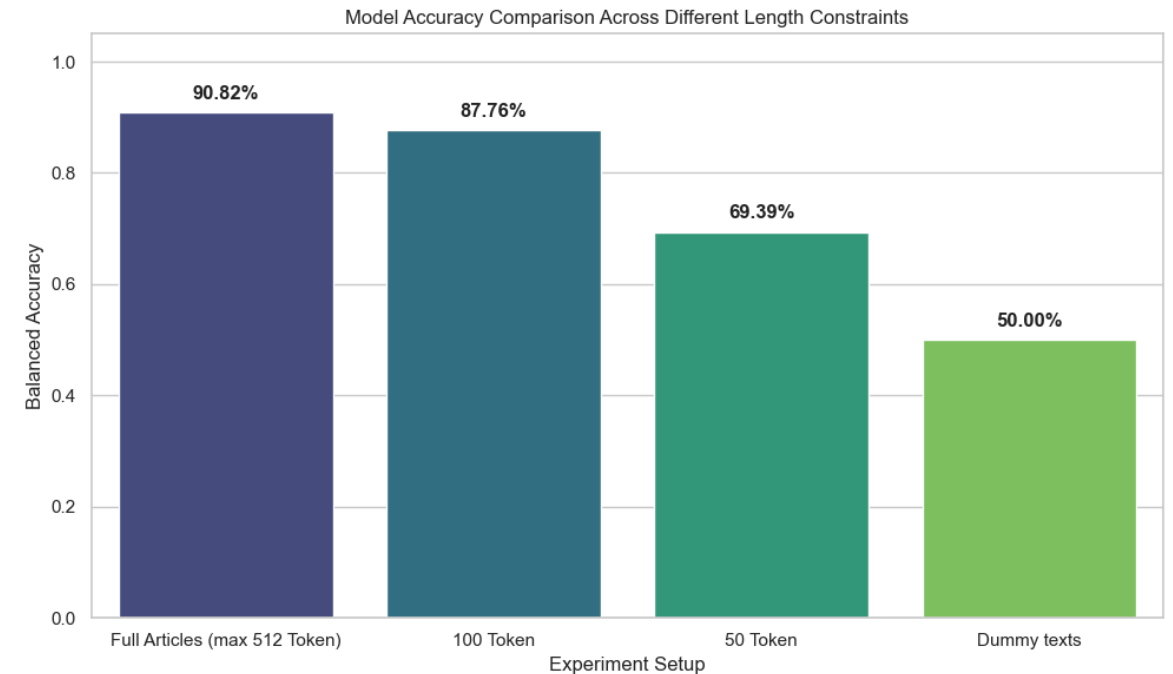
Content vs. Length Controls

Dummy Content Test

- All words replaced by a neutral token (▯), preserving original text lengths and padding patterns.
- **Balanced Accuracy: 50.0%** (classifies all as LS).

Constant-Length Slicing

- All texts sliced to exactly 50 or 100 tokens, eliminating all length variance.



Weekly Focus: Out-of-Domain Evaluation (Lebenshilfe)

- **Objective:** Verify the generalization capability of the BiLSTM classifier on a completely independent, unpublished dataset from *Lebenshilfe*.
- **Source Data:** 98 unstructured text documents (`.docx` , `.doc` , `.odt` , `.rtf`) containing manual translations into Easy Language (LS) and their everyday language (AS) counterparts.
- **Result:** A cleaned JSON dataset with **49 verified AS-LS article pairs** (98 documents total).

Out-of-Domain Classification Performance

We evaluated the BiLSTM models (trained on the similarity sweet spot `0.80 – 0.98`) on the 98 *Lebenshilfe* texts without fine-tuning:

Metric	Article-Level Model	Sentence-Level Model (Aggregated)	Sentence-Level Model (Sentence-Level)
Balanced Accuracy	90.82%	97.96%	79.71%
LS Correct (Simple)	93.88% (46/49)	97.96% (48/49)	76.02% (5877/7731)
AS Correct (Normal)	87.76% (43/49)	97.96% (48/49)	83.41% (1961/2351)

Sentence-level aggregation is performed via majority voting on the sentences of each article.

Next Steps & Future Work

- **Regression:**

- *Concept:* Map the degree of simplification between standard (AS) and simple language (LS) texts.
- *Variant 1 (Mix-Up):* Create artificial mixed sentences of equal length from LS and AS to study step-wise transition boundaries.
- *Variant 2 (LLM-in-the-Loop):* Feed AS and LS text pairs into LLMs as input to predict and measure progression levels between different simplification stages.

Week 14

Weekly Focus: Implementing the Regression Model

- **Variant 1 (Mix-Up):** Create artificial mixed sentences of equal length from LS and AS to study step-wise transition boundaries.
- **Variant 2 (LLM-in-the-Loop):** Feed AS and LS text pairs into LLMs as input to predict and measure progression levels between different simplification stages.

Variant 2: LLM-based Generation of Synthetic Intermediate Steps

- **Objective:** Establish a pipeline for generating progressive intermediate steps of text simplification between everyday language (AS) and Easy Language (LS) using LLMs.
- **Implementation:** Created a dedicated text generation script.
 - **Inputs:** Paired articles from the Lebenshilfe dataset.
 - **Outputs:** Progressively simplified versions stored incrementally in a new JSON dataset.
 - **Configurable Steps:** Supported generation of any float steps between `0.0` (LS) and `1.0` (AS), defaulting to `0.25`, `0.50`, and `0.75`.
 - **Features:** Robust resume functionality (detects existing output and skips completed steps), input limit option, and post-processing/cleaning of LLM output.

Test Run & Lessons Learned (Ollama & Remote Endpoint)

- **Local Execution:** local execution using Ollama and Llama 3.
- **Remote Execution:** Server can be pinged successfully, but the model itself cannot be invoked.
- **Key Issues Identified:**
 - i. **Model Chatty Prefixes:** LLMs frequently prepend introductory text (e.g., *"Hier ist der Text..."*), necessitating an enhanced post-processing script logic.
 - ii. **Layout Loss:** Easy Language formatting (line breaks, bullets) was lost in lower intermediate steps (e.g., 0.25), requiring prompt refinement.

Implementation Design for Variant 1 (Mix-Up)

Approach B (Sentence-Level Mixup):

Build paragraphs mixing AS and LS sentences (e.g. 50% LS sentences, 50% AS sentences for step 0.50).

- **Question:** Which sentences should be selected to ensure the content still makes sense?

Week 15

Weekly Focus: Implementing & Evaluating the Regression Setup

- **Goal:** Realization of the regression approaches (Mix-Up and LLM-based synthetic steps) to predict continuous complexity scores $\lambda \in [0.0, 1.0]$.
- **Approach 1 (Sentence-Level Mix-Up):** Implementation of the sentence-blending logic (First Variant), generation of training paragraphs, and analysis of target distribution.
- **Approach 2 (LLM Step Generation):** Running step-wise text generation, resolving LLM formatting/prefix issues, and performing dataset alignment cleanup.

Approach 2: LLM Step Generation Execution

- **Pipeline Run:** Executed `generate_synthetic_regression_steps.py` on the remote GPU server (`FlensGen-GPT-05S120B` model via VPN) and locally (LLaMA 3 via Ollama).
Remote Execution: Server can be pinged successfully, but the model itself cannot be invoked.

Approach 1: Mix-Up (First Variant)

First Variant Implementation:

- Sentence-level segmentation of parallel articles (LS & AS) using `spaCy`.
- Slices of randomly selected contiguous sentence ranges extracted independently from LS and AS and shuffled to build a paragraph.
- **Regression Target:** Calculated dynamically as the character length ratio of the LS portion to the total paragraph length.
- **Alternative Concept (Variant 2):** If the resulting target distribution of the first variant is not uniform enough for the regression model, a target $\lambda \sim U(0.0, 1.0)$ can be pre-sampled and sentence counts calculated accordingly.

Pseudocode (First Variant):

```
# Segment sentences
sents_ls = sentencize(ls_text)
sents_as = sentencize(as_text)

# Slice random contiguous blocks
start_ls, end_ls = rand_range(len(sents_ls))
start_as, end_as = rand_range(len(sents_as))
sample_ls = sents_ls[start_ls:end_ls]
sample_as = sents_as[start_as:end_as]

# Shuffle and calculate target
mixed = shuffle(sample_ls + sample_as)
target = char_len(sample_ls) / (
    char_len(sample_ls) + char_len(sample_as)
)
```

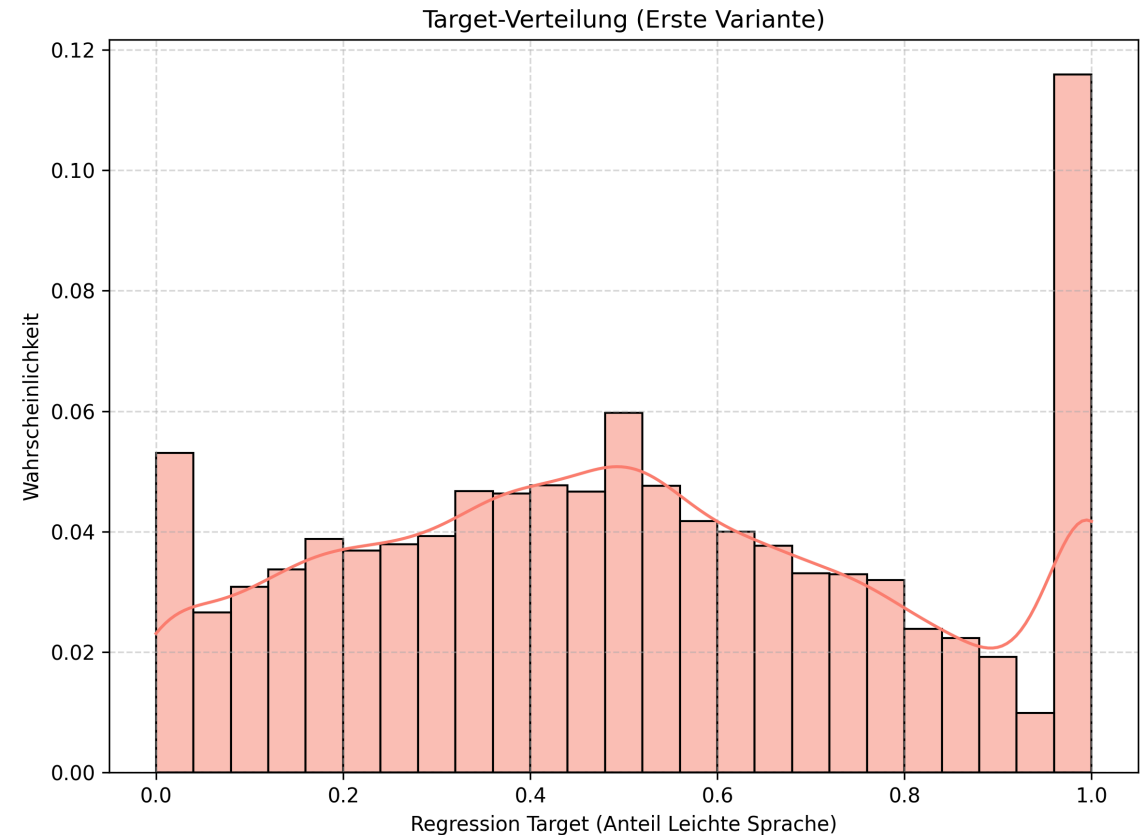
Target Distribution of the First Mix-Up Variant

- **First Variant Distribution:**

- The target distribution (peaking at 0.5 and near the boundaries) is usable for training, as the regression model should be robust enough.

- **Backup Concept (Variant 2):**

- Should the target distribution of the first variant lead to imprecise predictions at the extremes, Variant 2 is available as a backup concept where a uniformly distributed $\lambda \sim U(0.0, 1.0)$ is pre-sampled.



Next Steps & Research Questions

1. **Regression Training:** Implement model training based on the Mix-Up dataloaders.
2. **Remote Model Access:** Resolve the remote GPU server model invocation issue (server is pingable, but API does not return completions).

Week 16

Weekly Focus: Regression Training & Mix-Up Evaluation

- **Goal:** Train the regression model on the sentence-level Mix-Up approach.
- **Mix-Up Data Inspection:** Analyzed blended paragraph structure and manually checked calculated target values.
- **Model Stabilization:** Moved from an initial unstable setup (non-deterministic) to a fully converged, deterministic training pipeline.

Mix-Up Paragraph Generation & Target

- **DataLoader Logic:** Randomly extracts contiguous blocks from the Easy Language (LS) and Standard Language (AS) versions of an article, shuffles them, and computes the target.
- **Target (λ) Definition:** Character length ratio of the LS portion:

$$\lambda = \frac{\text{Length}(LS)}{\text{Length}(LS) + \text{Length}(AS)}$$

- **Linguistic Coherence:** Shuffling destroys logical paragraph coherence, which prevents the model from overfitting on semantic cues and forces it to focus strictly on sentence and stylistic complexity.

Mix-Up Paragraph Example

Source Sentences (German Extrakt):

- **Easy German (LS)** ($n = 2$):
 - *"Die Beauftragten der Bundes-Regierung für die Belange von Menschen mit Behinderungen..."*
 - *"Was macht der Behindertenbeauftragte..."*
- **Standard German (AS)** ($n = 5$):
 - *"Inhaltsverzeichnis Video: Was macht..."*
 - *"zum Download: Video: Was macht..."*
 - *"Gesetzlicher Auftrag..."*

Blended & Shuffled Paragraph:

Die Beauftragten der Bundes-Regierung... zum
Download: Video... Gesetzlicher Auftrag... Was
macht der Behindertenbeauftragte...
Inhaltsverzeichnis Video...

Computed Regression Target (λ):

$$\lambda = \frac{\text{CharLen}(LS)}{\text{CharLen}(LS) + \text{CharLen}(AS)} \approx 0.2087$$

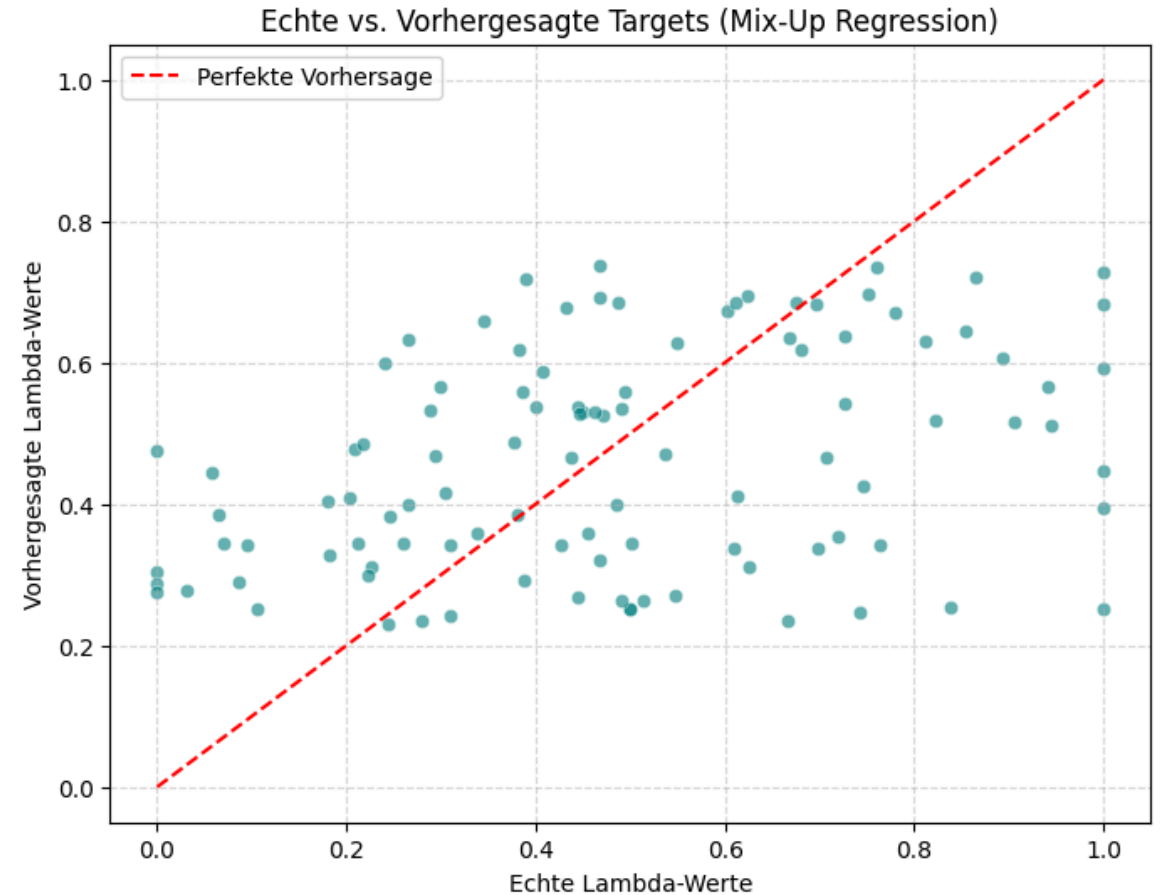
Initial Setup & Failure Mode (On-the-Fly Shuffling)

BiLSTM Regressor Setup:

- Embedding & Hidden Dim: 128, Dropout: 0.3, MSE Loss.
- Blending performed **on-the-fly** during `__getitem__`.

Problem & Diagnostic:

- Val MSE: `0.0655`, Val MAE: `0.2099` (deceptive).
- **Diagnosis:** Shuffling on-the-fly made the validation set non-deterministic every epoch.
- **Prediction Collapse:** Lacking a persistent sequence structure, the model predicted the mean (≈ 0.45).



Solution: Deterministic & Pre-Generated Datasets

- **Pre-generation:** The mixed paragraphs are generated once during dataset initialization rather than on-the-fly.
- **Reproducibility:** Seeded random generator (42 for train, 99 for validation) ensures identical samples across all epochs.
- **Data Augmentation:**
 - **Train:** 10 fixed mixes per article pair → **9,280 samples.**
 - **Validation:** 2 fixed mixes per article pair → **208 samples.**

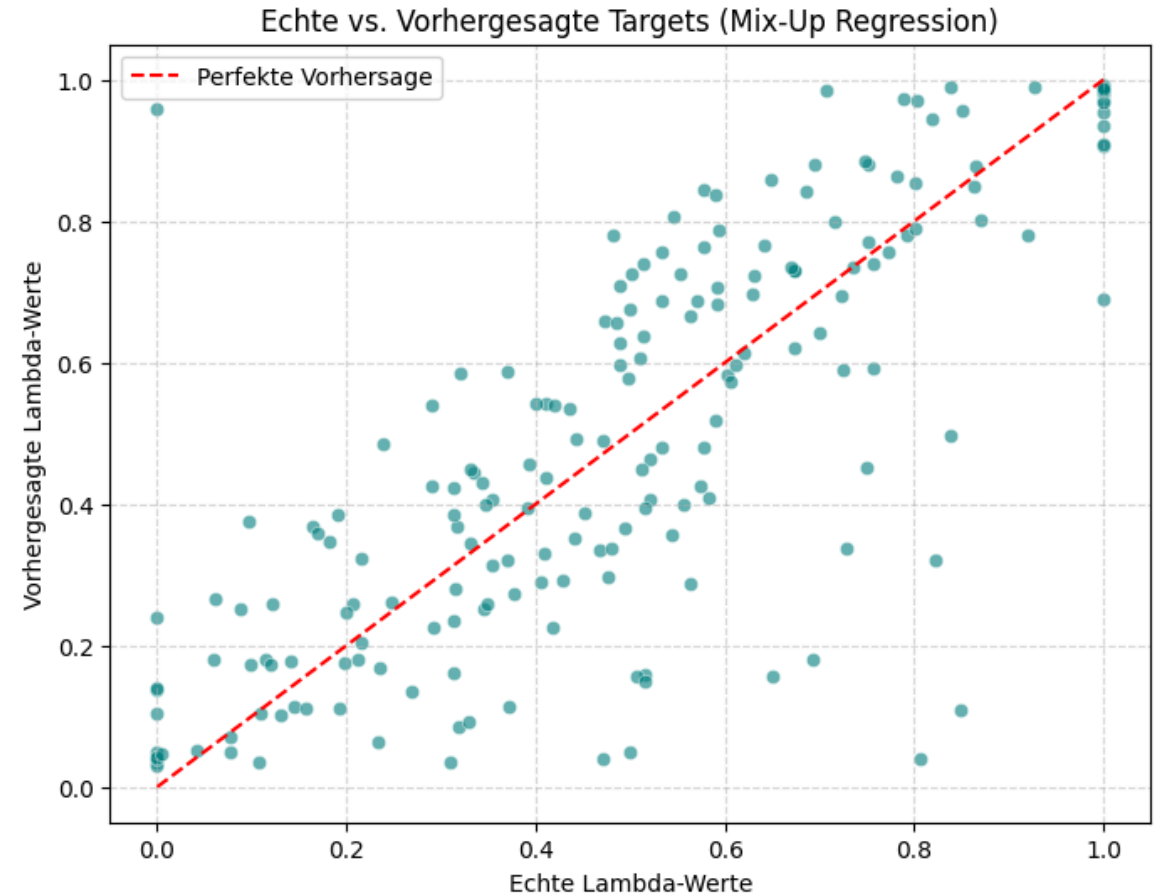
Final Results

Results at Epoch 23 (Early Stopping at 28):

- **Validation MSE: 0.0335** (halved from 0.0655)
- **Validation MAE: 0.1195** (halved from 0.2099)

Interpretation:

- **Continuous Curve:** Predictions align more with the diagonal $y = x$.
- **Extremes Accuracy:** Clean AS ($\lambda = 0$) and LS ($\lambda = 1$) are predicted with high confidence near their true values.



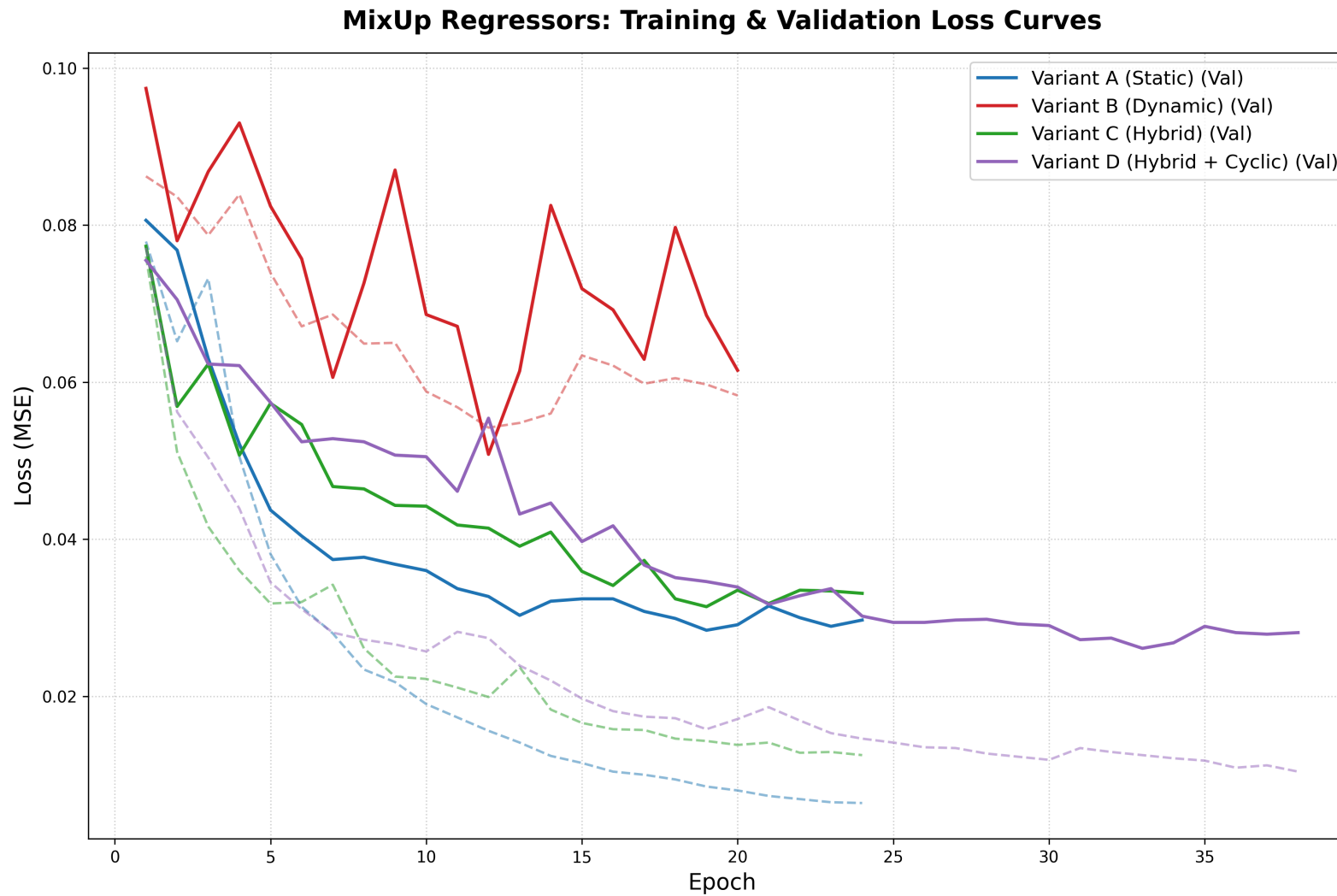
Next Steps

1. **Evaluate on Lebenshilfe dataset:** We should get only extremes?
2. **Evaluate LLM-Generated Levels (Approach 2):** Run the trained Mix-Up BiLSTM model on the synthetic LLM text stages (0.25 , 0.50 , 0.75) to assess if the predicted complexity aligns with prompt instructions.

Week 17

MixUp Regressors: Comparing the 4 Training Variants

- **Variant A: Static Pre-mixing**
 - *Concept:* Sentence mixing and shuffling are generated once during dataset initialization.
- **Variant B: Dynamic Mixing**
 - *Concept:* Random shuffling and mixing of sentences performed on-the-fly in `__getitem__`.
- **Variant C: Hybrid Solution**
 - *Concept:* Combination of static and dynamic. The probability of dynamic mixing $p_{dynamic}$ increases linearly across epochs from 0.0 to 1.0.
- **Variant D: Hybrid Solution + Cyclic LR**
 - *Concept:* Based on the Hybrid Solution (Variant C), but introduces a cyclic learning rate scheduler (`CosineAnnealingWarmRestarts`).



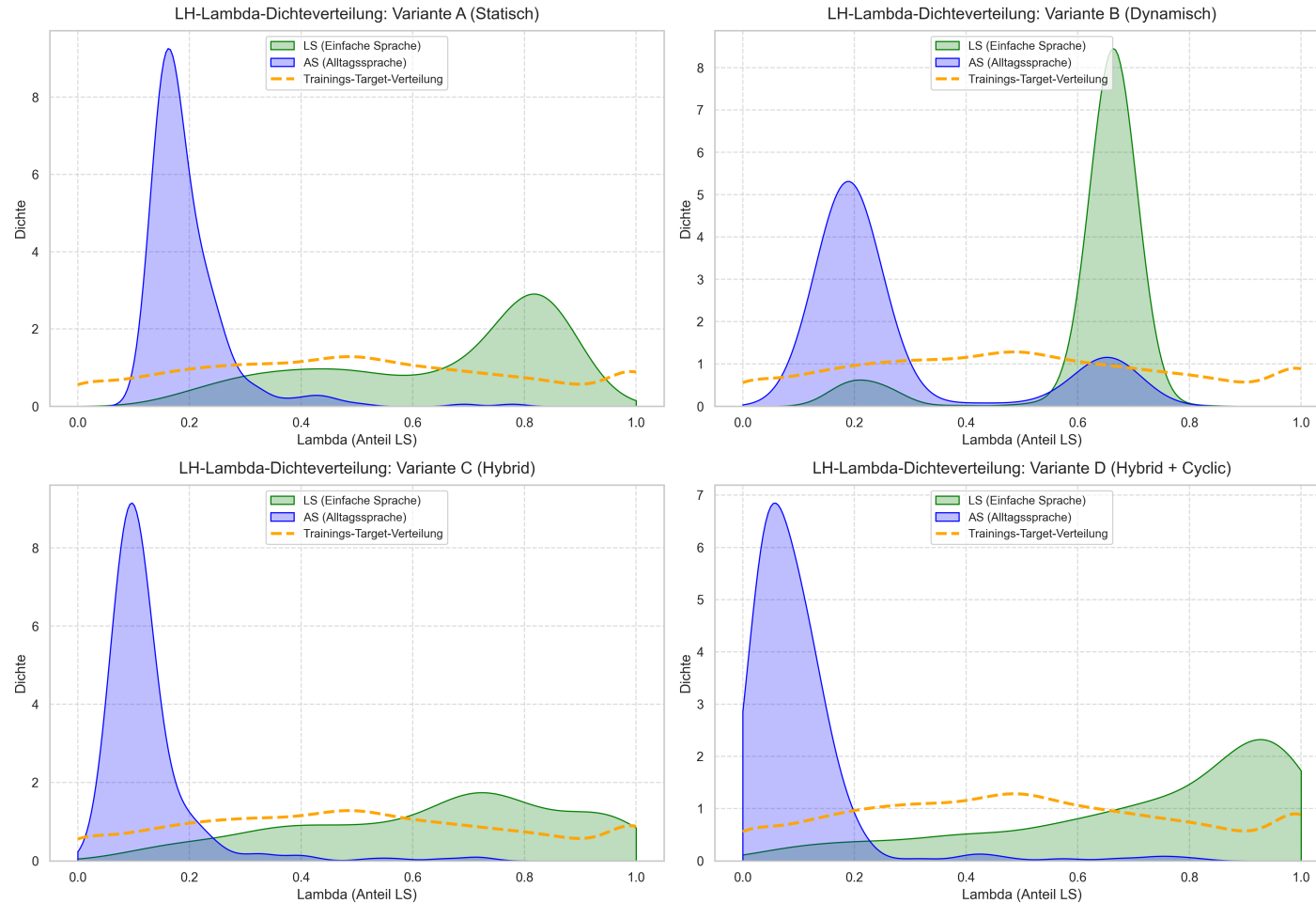
MixUp Regressors: Training & Validation Loss Curves

Evaluation on the Lebenshilfe Test Set

Model	$\emptyset \lambda$ (LS)	$\emptyset \lambda$ (AS)	Acc (0.5)	Balanced Acc	MAE (1/0)
A (Static)	0.6518	0.1176	87.16%	89.20%	0.2597
B (Dynamic)	0.5516	0.2811	77.41%	81.48%	0.3842
C (Hybrid)	0.6315	0.1323	83.18%	85.98%	0.2779
D (Hybrid + Cyclic)	0.7554	0.1051	91.78%	92.83%	0.1911

- **Best Results:** Variant D (Hybrid + Cyclic) dominates across all metrics.
- **Worst Results:** Variant B (Dynamic) struggles to learn stable representations.

Vergleich der MixUp-Regressoren auf dem Lebenshilfe-Datensatz (Reine Sätze) vs. Trainings-Target-Verteilung



Analysis of the Lebenshilfe KDE Plots

Future Optimization Steps (Backlog)

- **Train with LLM-Generated Levels:**
 - Train the model on synthetic text stages (0.25 , 0.50 , 0.75).
- **Variant B with Cyclic LR:**
 - Train the purely dynamic mixing model (Variant B) using a cyclic learning rate scheduler (e.g., Cosine Annealing with Warm Restarts) to see if periodic momentum helps it escape local minima and overcome its initial convergence plateau.

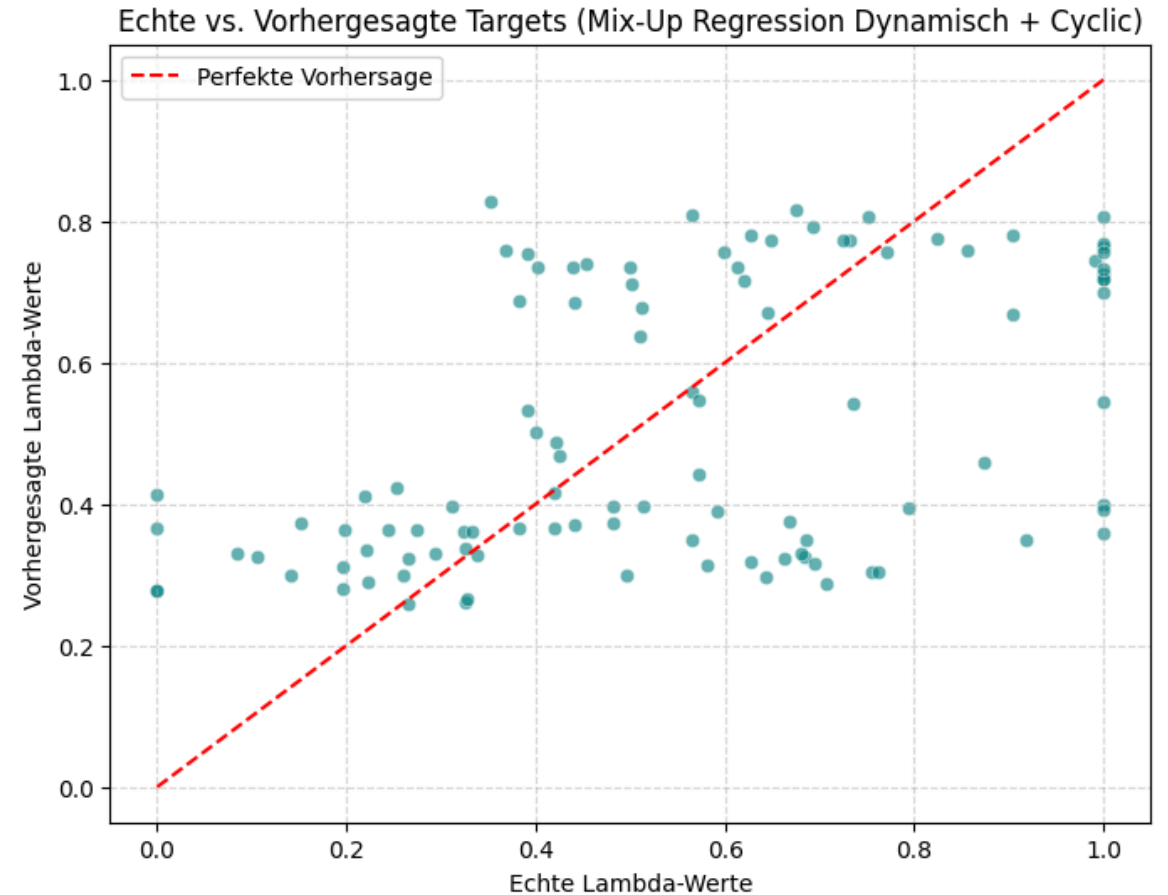
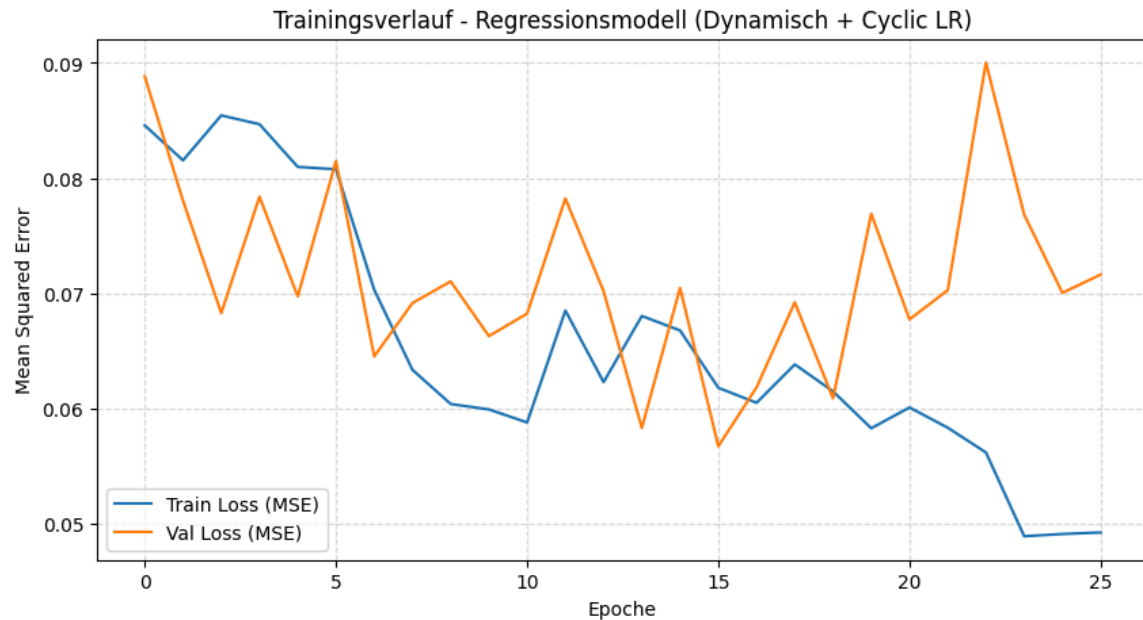
Week 18

Weekly Focus: MixUp Optimization, Thesis Summary & Translation Model Planning

- **1. MixUp Regressor Optimization & Evaluation:** Cyclic LR evaluation on Variant B, diagnostic error analysis, and complete In-Domain & Out-of-Domain (Lebenshilfe) performance comparison across all 4 variants (MixUp & Non-MixUp).
- **2. Master Thesis Status Summary & Overall Architecture:** High-Level Overview for status quo.
- **3. Planning Translation Model (Step 3):** Model selection (Seq2Seq vs. Causal LLMs) and reward guided training.

Part 1: MixUp Regressor Optimization & Evaluation

1.1 MixUp Variant B + Cyclic LR: Loss & Scatterplot Analysis



1.2 In-Domain Evaluation (Test-Split)

A. Continuous MixUp Regression (MixUp Evaluation)

Model Variant	Test MSE	Test MAE
Variant A (Static)	0.0383	0.1557
Variant B (Dynamic)	0.0758	0.2264
Variant D (Hybrid + Cyclic)	0.0241	0.1027

B. Binary Classification on Pure Sentences (Non-MixUp Evaluation: LS = 1.0, AS = 0.0)

Model Variant	$\emptyset \lambda_{LS}$	$\emptyset \lambda_{AS}$	Accuracy (0.5)	Balanced Acc	MAE (Target 1/0)
Variant A (Static)	0.7596	0.2680	91.55%	91.46%	0.2526
Variant B (Dynamic)	0.6138	0.3312	80.17%	79.53%	0.3620
Variant D (Hybrid + Cyclic)	0.9007	0.1382	95.92%	95.93%	0.1164

1.3 Out-of-Domain Evaluation (Lebenshilfe Dataset)

A. Continuous MixUp Regression (MixUp Evaluation)

Model Variant	LH MSE	LH MAE
Variant A (Static)	0.0725	0.2111
Variant B (Dynamic)	0.0766	0.2212
Variant D (Hybrid + Cyclic)	0.0739	0.2087

B. Binary Classification on Pure Sentences (Non-MixUp Evaluation: LS = 1.0, AS = 0.0)

Model Variant	$\emptyset \lambda_{LS}$	$\emptyset \lambda_{AS}$	Accuracy (0.5)	Balanced Acc	MAE (Target 1/0)
Variant A (Static)	0.6533	0.1958	82.67%	85.75%	0.2888
Variant B (Dynamic)	0.6241	0.2766	87.42%	86.32%	0.3378
Variant D (Hybrid + Cyclic)	0.7293	0.0957	87.29%	89.37%	0.2036

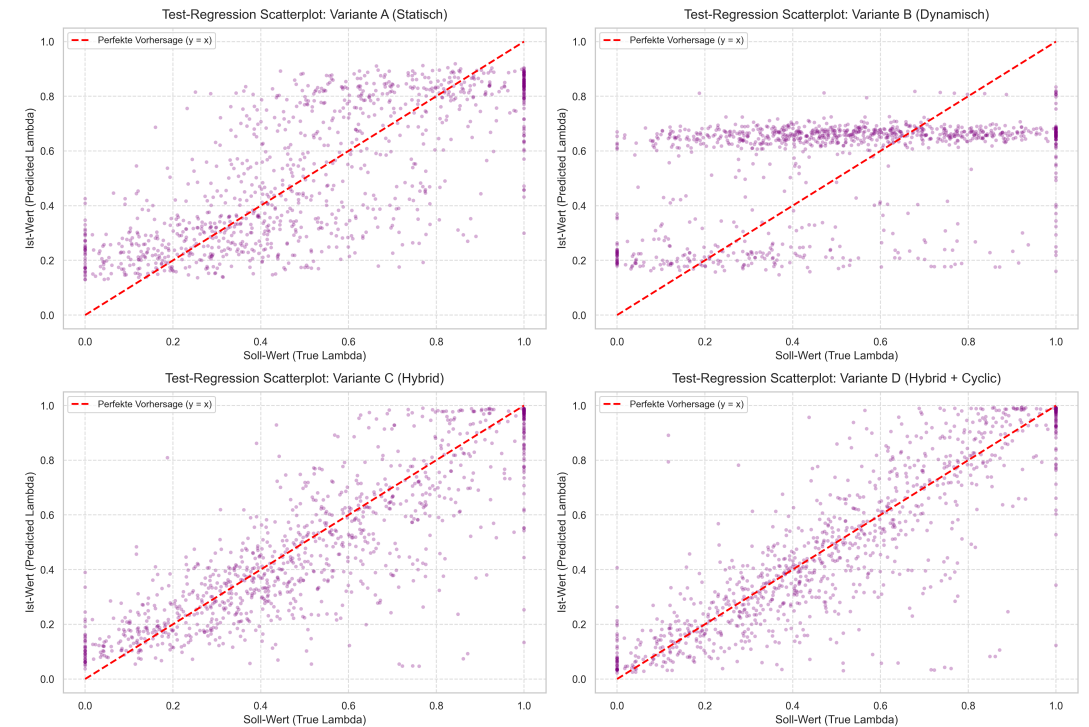
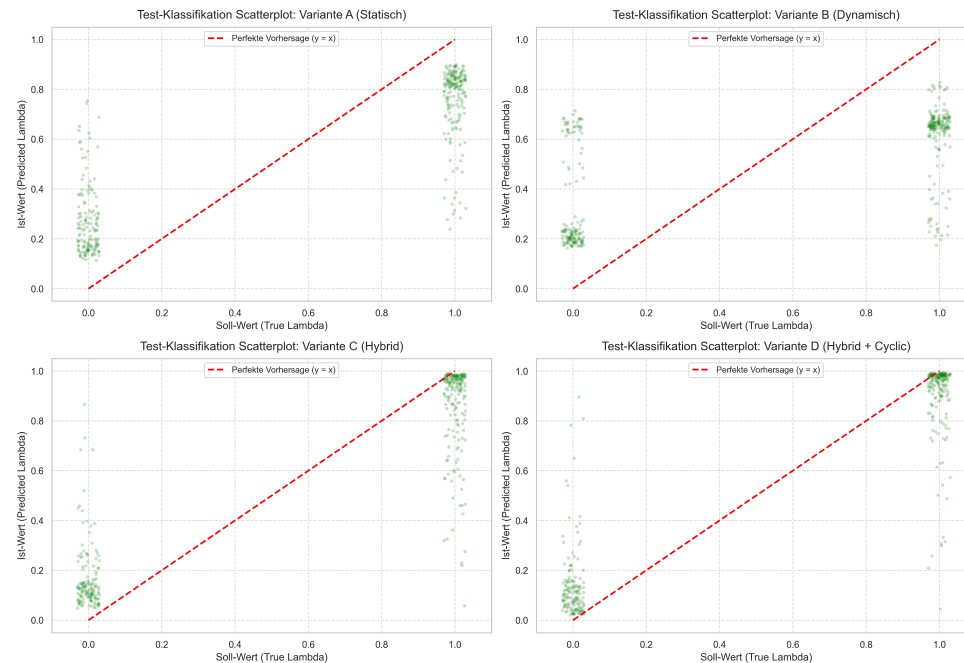
1.4 In-Domain Test-Split Scatterplots: Non-MixUp vs. MixUp

Binary Classification (Non-MixUp: Pure Sentences):

Continuous Regression (MixUp Blends):

Vergleich der MixUp-Regressoren: Ist- vs. Soll-Werte auf dem Test-Split

Vergleich der MixUp-Regressoren: Ist- vs. Soll-Werte auf dem Test-Split (Reine Sätze)

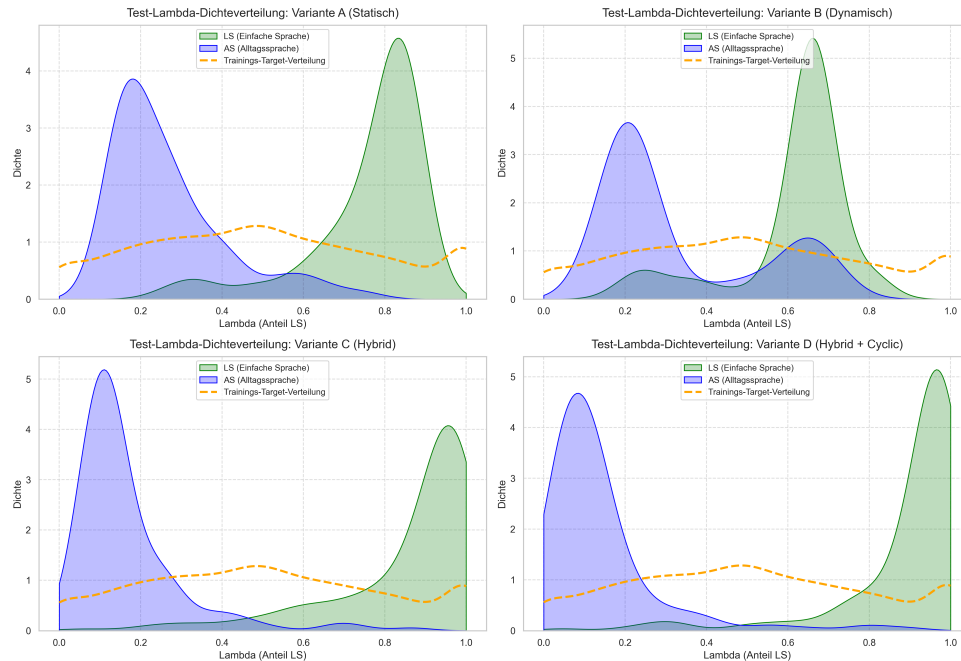


1.5 In-Domain Test-Split Density Distributions

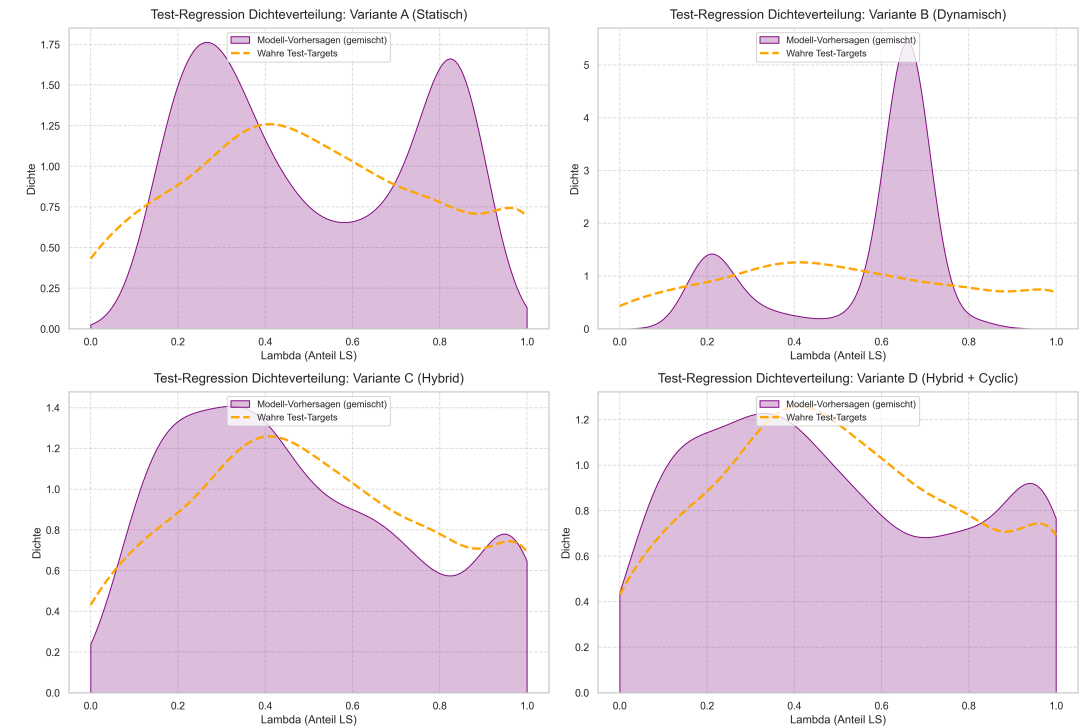
Binary Classification (Non-MixUp: Pure Sentences):

Continuous Regression (MixUp Blends):

Vergleich der MixUp-Regressoren auf dem Test-Split (Reine Sätze) vs. Trainings-Target-Verteilung



Vergleich der MixUp-Regressoren: Vorhersage- vs. Target-Verteilung auf dem gemischten Test-Split



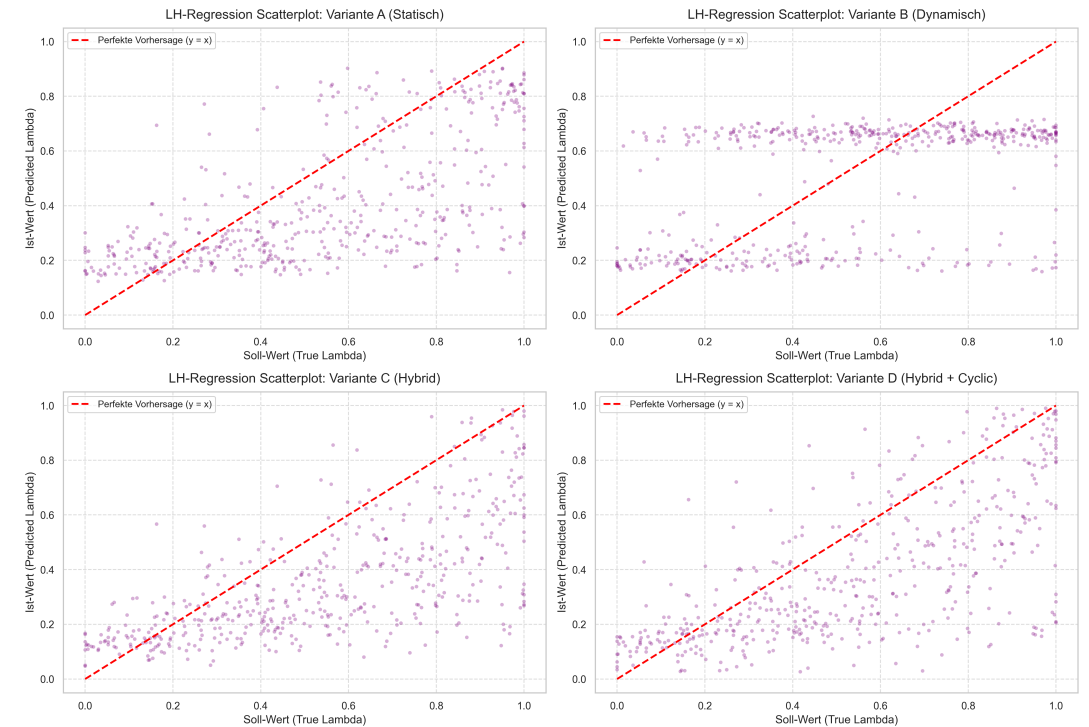
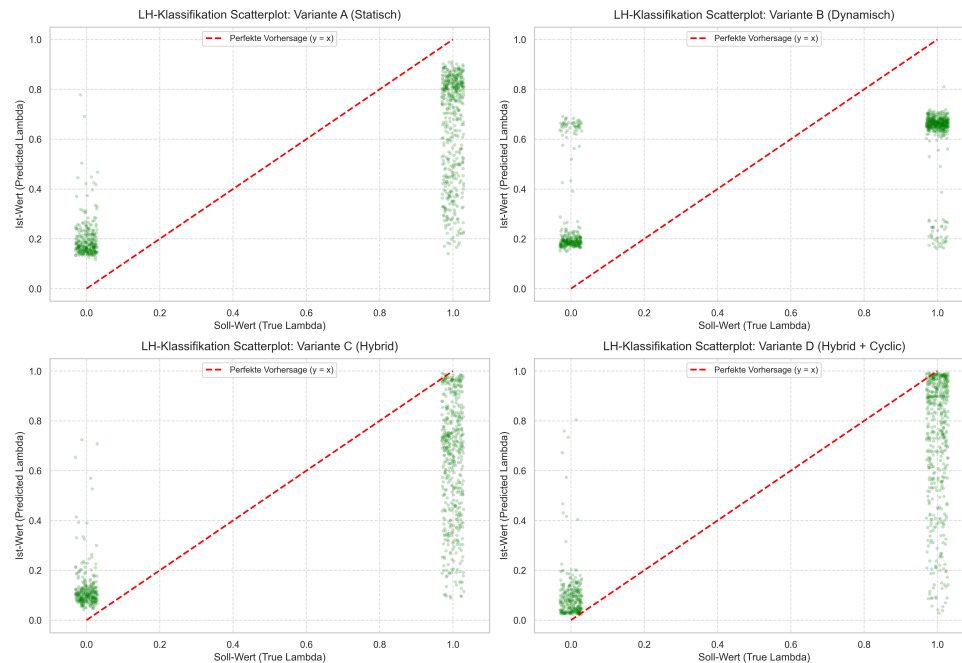
1.6 Out-of-Domain Lebenshilfe Scatterplots: Non-MixUp vs. MixUp

Binary Classification (Non-MixUp: Pure Sentences):

Continuous Regression (MixUp Blends):

Vergleich der MixUp-Regressoren: Ist- vs. Soll-Werte auf dem Lebenshilfe-Set

Vergleich der MixUp-Regressoren: Ist- vs. Soll-Werte auf dem Lebenshilfe-Set (Reine Sätze)

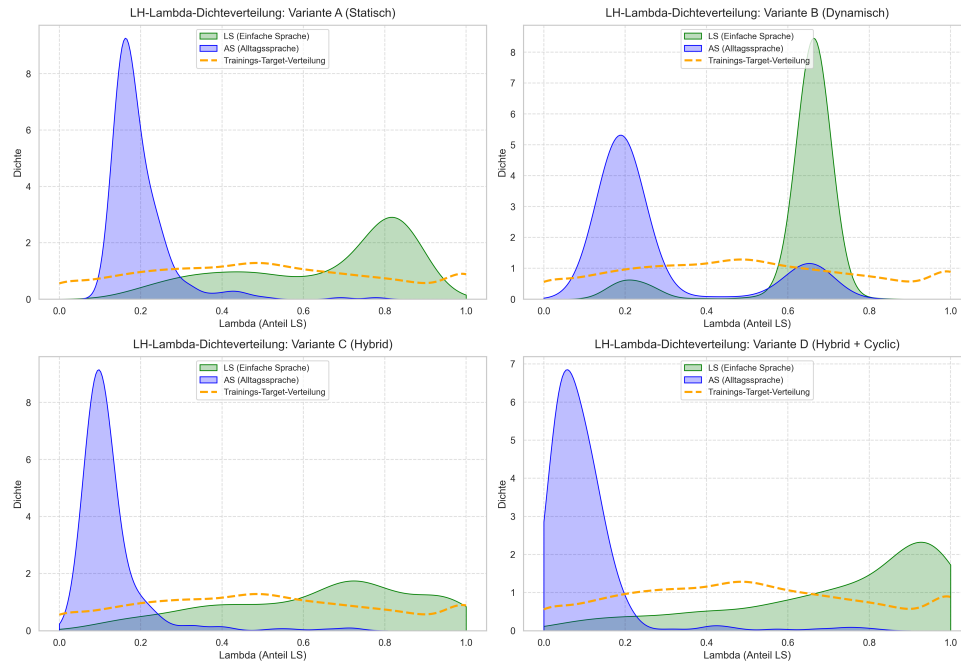


1.7 Out-of-Domain Lebenshilfe Density Distributions

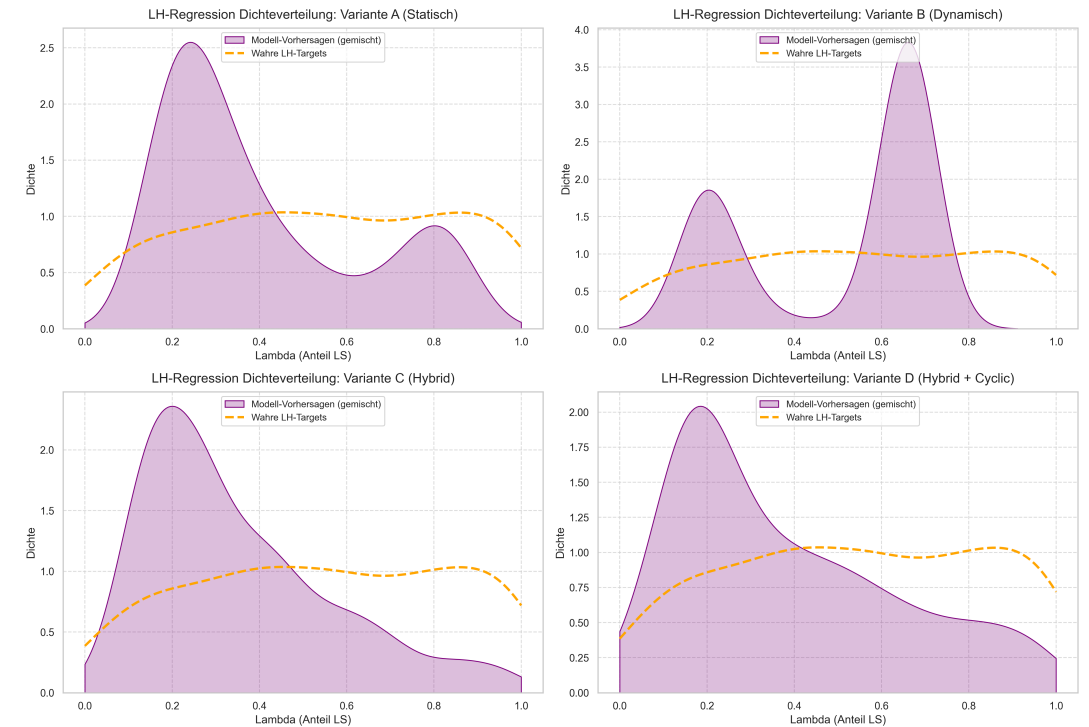
Binary Classification (Non-MixUp: Pure Sentences):

Continuous Regression (MixUp Blends):

Vergleich der MixUp-Regressoren auf dem Lebenshilfe-Datensatz (Reine Sätze) vs. Trainings-Target-Verteilung

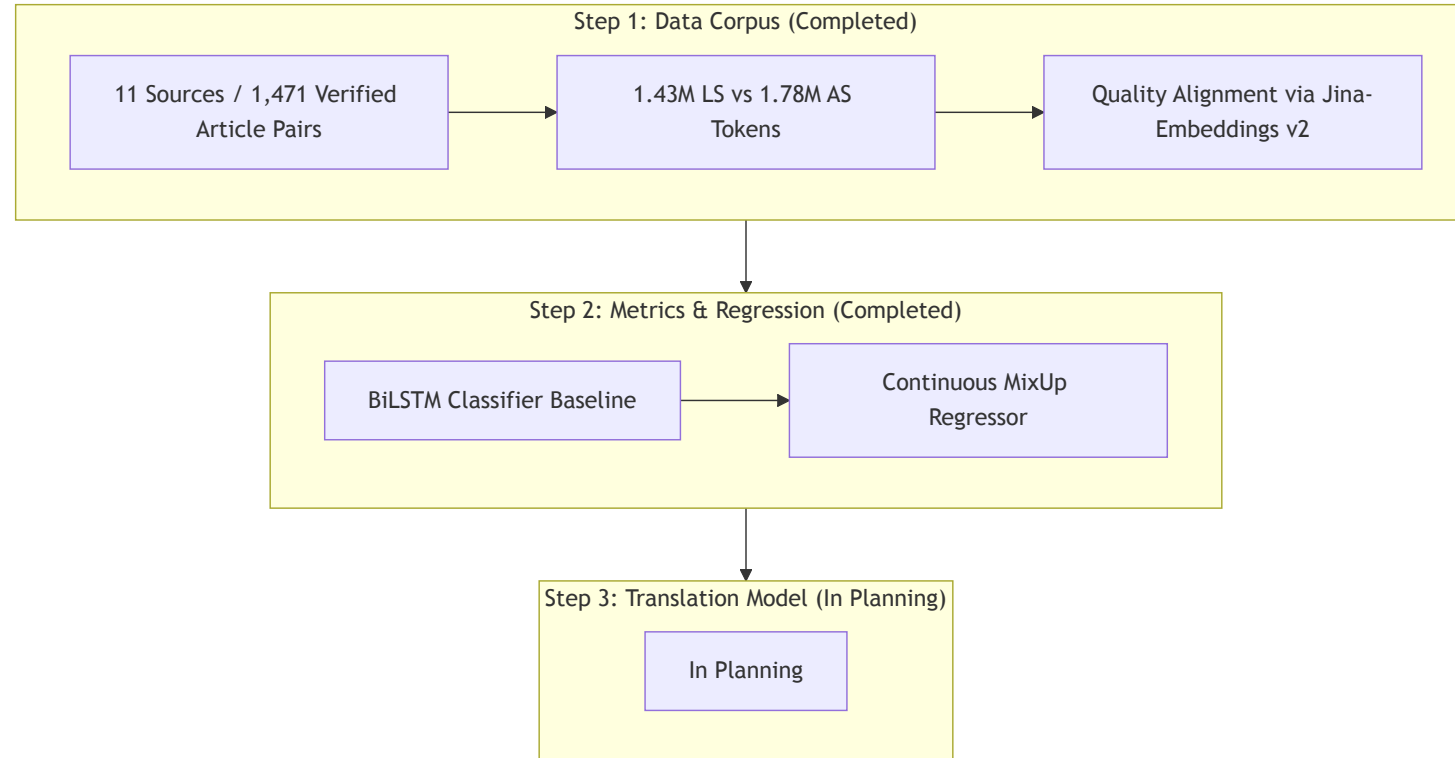


Vergleich der MixUp-Regressoren: Vorhersage- vs. Target-Verteilung auf dem gemischten Lebenshilfe-Set



Part 2: Thesis Status Summary & Big Picture

2.1 Big Picture: 3-Step Master Thesis Architecture



2.2 Step 1 Status: Data Corpus & Quality Assurance

Corpus Statistics (Final Cleaned State):

- **1,471** verified article pairs across 11 sources.
- **1.43M Easy German Tokens** vs. **1.78M Standard German Tokens**.
- **91,103 LS Sentences** vs. **54,256 AS Sentences**.

Quality Assurance Pipeline:

- **Long-Context Alignment:** `jina-embeddings-v2-base-de` (8,192 token window) for section-level matching.
- **Similarity Sweet-Spot:** Filtered range $0.80 \leq \text{Similarity} \leq 0.98$.

Top Sources	Pairs	Tokens (LS)	Tokens (AS)
Hannover.de	796	861,967	858,086
MDR	227	94,976	168,440
Apotheken Umschau	157	234,608	443,722
Hamburg.de	56	61,204	66,977
Behindertenbeauftragter	51	38,724	44,725
Wiesbaden.de	41	13,808	23,332
Lebenshilfe (Test)	34	9,974	11,037
TOTAL (Final)	1,471	1,429,433	1.781.714

2.3 Step 2 Status: Classification & MixUp Regression

1. Binary Classification Baseline Models

Model Architecture	Scope / Aggregation	In-Domain BAcc	Out-of-Domain BAcc (LH)
Sentence BiLSTM (Maj. Vote)	Sentence → Article	99.68%	96.94%
Article BiLSTM Baseline	Article (512 tokens)	99.03%	90.82%
Sentence BiLSTM (Raw)	Single Sentence	92.99%	78.76%

2. Continuous MixUp Regressor Variants

Model Variant	Training Strategy	Test MSE	Test MAE	In-Domain BAcc	Out-of-Domain BAcc (LH)
Variant A	Static (Pre-mixed)	0.0383	0.1557	91.46%	85.75%
Variant B	Dynamic (<code>__getitem__</code>)	0.0758	0.2264	79.53%	86.32%
Variant C	Hybrid Schedule	0.0267	0.1158	95.22%	85.10%
Variant D	Hybrid + Cyclic LR	0.0241	0.1027	95.93%	89.37%

2.4 Project Progress & Milestone Dashboard

Milestone	Status	Key Metrics & Achievements
1. Web Crawling & Corpus Building	Completed	1,471 Pairs, 1.43M Tokens, 11 Sources
2. Quality & Embedding Alignment	Completed	Jina-Embeddings-v2 (8k), Similarity 0.80–0.98
3. BiLSTM Classifier Baseline	Completed	99.68% Balanced Accuracy (Majority Vote)
4. Continuous MixUp Regressor	Completed	MSE 0.0241, MAE 0.1027 (Variant D)
5. Synthetic LLM Levels Eval	In Progress	Ollama / FlensGen-GPT Evaluation
7. Translation	Planning	mBART-50 & LLaMA-3-8B (LoRA/QLoRA) & Reward

Part 3: Planning Translation Model (Step 3)

3.1 Step 3: Model Architectures & Approaches

A. Sequence-to-Sequence (Encoder-Decoder)

- **Models:** `mBART-50` (`facebook/mbart-large-50`), `mt5` (`google/mt5-base` / `mt5-large`).

B. Causal LLMs (Decoder-Only via LoRA / QLoRA)

- **Models:** `LLaMA-3-8B-Instruct` , `Mistral-7B-v0.3` , `Qwen2.5-7B` .

Metric Models as Reward Function

- Reinforcement Learning / Direct Preference Optimization / Metric Guidance

Open Questions & Discussion

- **Reward Function:**
 - Proceed with a metric model (the best one) as the reward function?
- **Dataset Usage:**
 - Is it valid to train/evaluate multiple models (metric & translation models) on the same dataset?

Week 19

Weekly Focus: Thesis Structure, Synthetic Regression Evaluation & Translation Model SFT

- **1. Master Thesis Structure:** Overall outline of chapters 1-7.
- **2. Synthetic LLM-based Regression vs. MixUp:** Implementation of `FlensGen-GPT-0SS-120B` generated continuous complexity steps, evaluation of `BiLSTMRegressor` on synthetic steps, and comparative analysis.
- **3. Translation Model (Step 3):** Supervised Fine-Tuning (SFT) pipeline, baseline results on `google/mt5-small` with decoding constraints, and DPO training progress.

Part 1: Master Thesis Structure Outline

1.1 Thesis Structure: Chapters 1-4

1. Einleitung

- 1.1 Problemstellung & Motivation (Barrierefreiheit)
- 1.2 Zielsetzung & Forschungsfragen
- 1.3 Aufbau (Drei-Stufen-Pipeline)

2. Background & Stand der Forschung

- 2.1 Leichte Sprache (Regeln, Zielgruppen)
- 2.2 MT & Textvereinfachung (Seq2Seq, LLMs)
- 2.3 Evaluierungsansätze (Metriken, DPO)

3. Themenblock 1: Korpus-Erstellung

- 3.1 Multi-Quellen-Crawling (11-Quellen-Korpus)
- 3.2 Alignment & Das 1:n-Problem (Satz/Absatz)
- 3.3 Quality Assurance (Similarity Sweet-Spot)
- 3.4 Datenbereinigung & Normalisierung
- 3.5 Korpus-Statistiken (Diversität, TTR)

4. Themenblock 2: Sprachkomplexität

- 4.1 Klassifikationsmodelle (BiLSTM vs. SBERT)
- 4.2 Out-of-Domain-Generalisierung & Bias
- 4.3 Continuous MixUp Regression
- 4.4 Evaluierung synthetischer Sprachstufen

1.2 Thesis Structure: Chapters 5-7

5. Themenblock 3: Übersetzung & Reward-Guided Fine-Tuning

- 5.1 Datenvorbereitung & Trainings-Splits (Gefilterter Sweet-Spot, Block-basiert)
- 5.2 Supervised Fine-Tuning (SFT) (mt5 vs. Mistral/LLaMA)
- 5.3 Reward-Guided Optimization (RLHF / DPO via MixUp-Regressor Reward)
- 5.4 Mehrdimensionale Evaluierung (Regelkonformität, NLI/NER Faktentreue)

6. Diskussion & Gesamtevaluation

- 6.1 Inhaltssynthese der drei Themenblöcke
- 6.2 Stärken, Grenzen & Systemrestriktionen
- 6.3 Beantwortung der Forschungsfragen

7. Fazit & Ausblick

- 7.1 Zusammenfassung der Kernergebnisse
- 7.2 Ausblick & Zukünftige Arbeiten

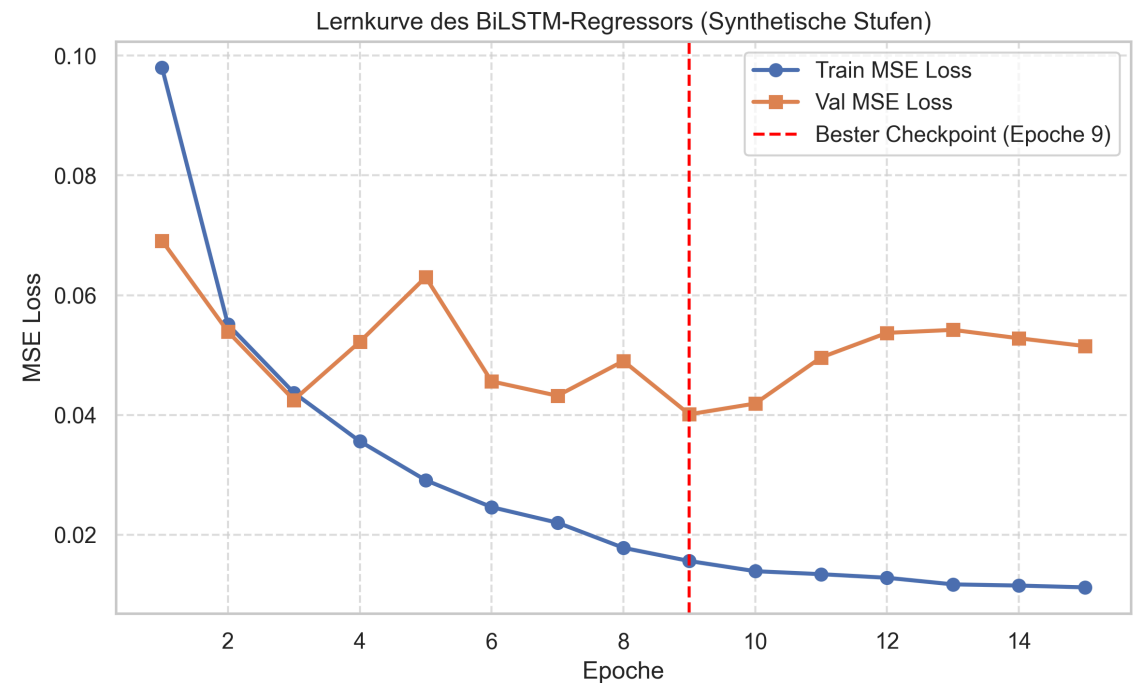
Part 2: MixUp vs. Synthetischer Regressor

2.1 Synthetische Datengenerierung & Modelltraining

- **Ziel:** Generierung semantisch konsistenter und grammatikalisch flüssiger Zwischenstufen (`0.25` , `0.50` , `0.75`) mit `FlensGen-GPT-05S-120B` .
- **Erzeugte Datensätze:**
 - **LH-Eval:** `lebenshilfe_dataset_with_steps.json` (245 Samples)
 - **Haupt-Korpus:** `corpus_final_with_steps.json` (7.380 Samples)
- **Modelltraining (`1_synthetic_bilstm_regression.ipynb`):**
 - Trainierte `BiLSTMRegressor` auf ausgeflachten Text-Target-Paaren.
 - Validation MSE erreichte Minimum von `0.0401` in Epoche 9.
 - **LH Out-of-Domain Generalisierung:** MSE: `0.0195` | MAE: `0.1042` | Pearson r : `0.9238` | Spearman ρ : `0.9169` .

2.2 Lernkurve: Synthetischer BiLSTM Regressor

- Der Regressor lernt stabil und konvergiert schnell.
- Hervorragende Generalisierung auf den ungesehenen Lebenshilfe-Datensatz mit hoher linearer und Rangkorrelation (>0.91).



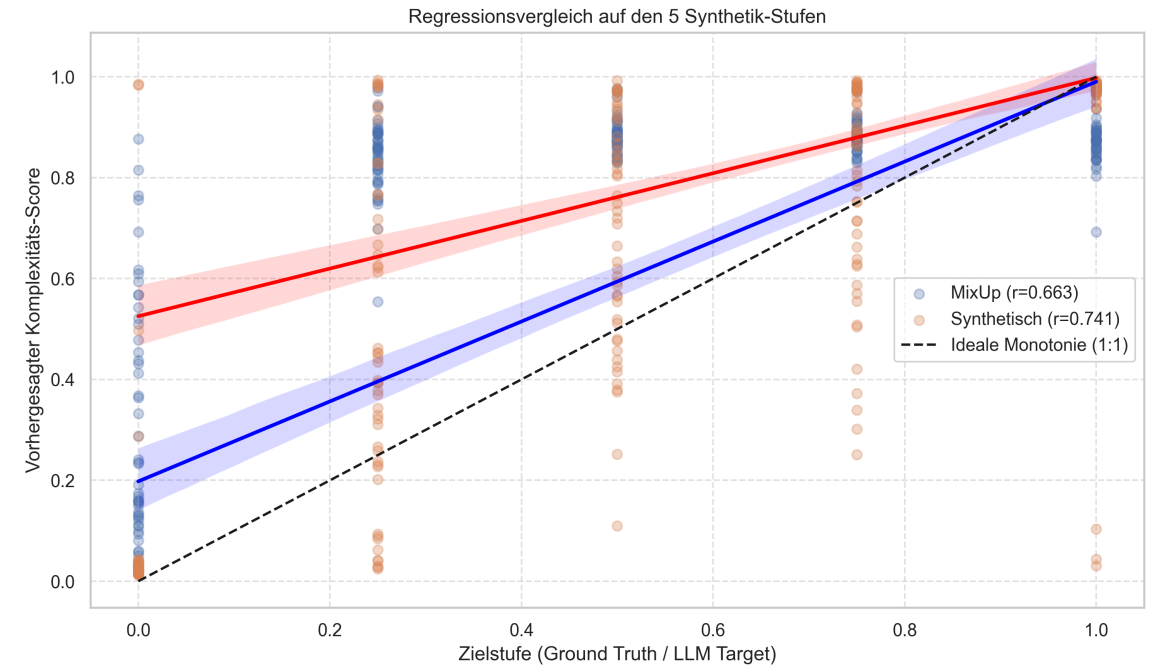
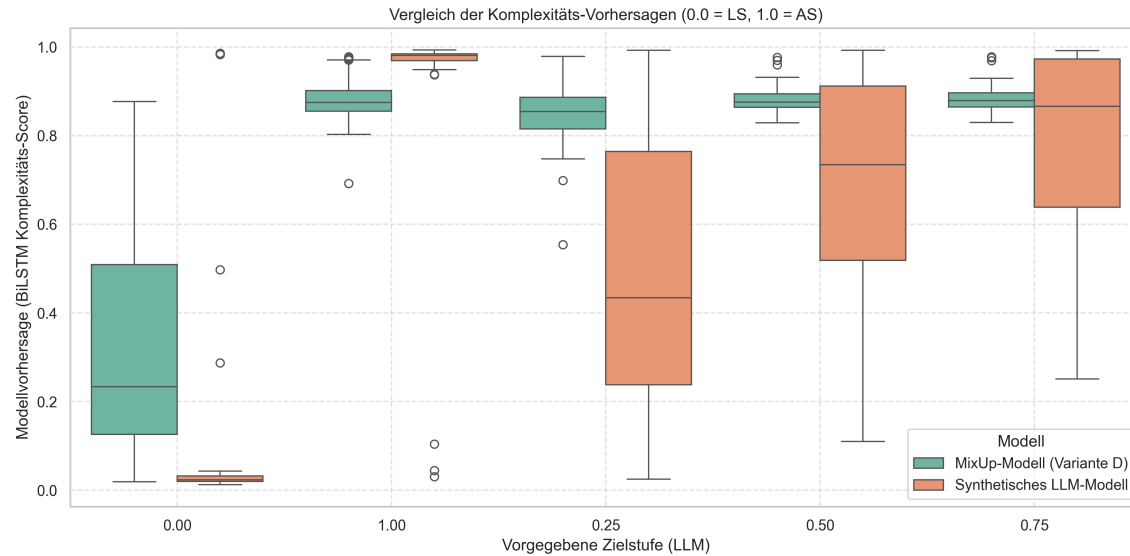
2.3 Systematischer Vergleich: MixUp vs. Synthetisches LLM-Modell

- **Skalen-Harmonisierung:** MixUp-Vorhersage harmonisiert auf Komplexitäts-Skala ($1.0 - \text{Pred}$), so dass $0.0 = \text{LS}$ und $1.0 = \text{AS}$.
- Evaluation auf dem exakt selben Test-Set (Lebenshilfe mit 5 Stufen):

Metrik	MixUp-Modell (Variante D)	Synthetisches LLM-Modell	Gewinn durch Synthetik-Ansatz
MSE	0.1388	0.0786	-43,4 %
MAE	0.3079	0.1816	-41,0 %
Pearson r	0.6626	0.7412	+11,9 %
Spearman rho	0.5885	0.7391	+25,6 %

Ergebnis: Synthetische Stufen aus dem LLM spiegeln echte Komplexität wider. Das synthetische Modell wird als **Reward-Modell** in Step 3 integriert.

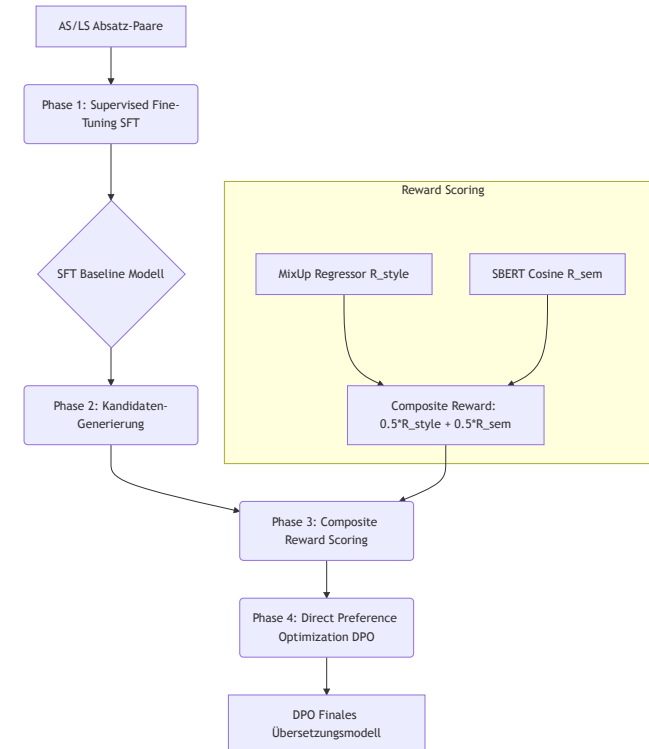
2.4 Boxplot- & Regressions-Vergleich



Part 3: Translation Model: Seq2Seq SFT & DPO

3.1 Pipeline-Design & Trainingsschritte

- **Basismodell:** google/mt5-small (~300M Parameter).
- **Trainingsdaten:** 1.471 aligned Absatz-Paare (85% Train / 15% Val).
- **Out-of-Domain Test:** LH-Datensatz (49 Artikel-Paare).
- **Phase 1:** Supervised Fine-Tuning (SFT) mittels standardmäßigem Cross-Entropy-Loss.
- **Phase 2 & 3:** Generierung von Kandidaten und Bewertung über Composite Reward:
$$R = 0.5 \cdot R_{\text{style}} + 0.5 \cdot R_{\text{sem}}$$
- **Phase 4:** Direct Preference Optimization (DPO).



3.2 SFT Baseline-Ergebnisse (Lebenshilfe Testset)

- **Ergebnis 1: SFT 1 Epoche (Greedy, keine Constraints)**
 - Train Loss: 9.3095 . Extrem untertrainiert, endlose Wiederholungsschleifen.
- **Ergebnis 2: SFT 5 Epochen (mit Decoding-Constraints)**
 - Train Loss: 3.2545 . Gute Konvergenz. Wiederholungen eliminiert durch Beam Search und Repetition Penalties.
 - Bildet selbstständig kurze Sätze und Fragesätze.

Metrik	Ergebnis 1 (1 Epoche)	Ergebnis 2 (5 Epochen)	Verbesserung
Ø MixUp-Score (R_{style})	0.7551	0.8105	+5.5%
Ø SBERT-Similarity zur AS-Quelle (R_{sem})	0.7663	0.8773	+11.1%
Ø SBERT-Similarity zur LS-Referenz	0.7354	0.8428	+10.7%
Ø Composite Reward	0.7607	0.8439	+8.3%

3.3 DPO Trainings-Status

- Das 5-Epochen SFT-Modell wurde unter `results/best_sft_model_temp.pt` gesichert.
- **Optimierungen gegen GPU-VRAM OOM (8 GB VRAM):**
 - i. Offloading von SBERT und BiLSTM-Regressor auf die **CPU** (VRAM-Ersparnis >1 GB).
 - ii. Reduktion der DPO-Batch-Größe auf `batch_size = 2`.
- **Status:** DPO-Training läuft aktuell. Ergebnisse des Tunings und der finalen Übersetzungsevaluierung folgen im nächsten Meilenstein.